PCEOT402
Electrical Machines
Module 3 — Synchronous Machines
Complete Study Notes from Parts 1, 2 & 3

1. Principle of operation of the 3-phase synchronous generator
2. Classification and constructional features of alternators
3. Speed of rotation, pole relation and generated frequency
4. Pitch factor, distribution factor, winding factor and EMF equation
5. Open-circuit characteristics and no-load operation
6. Armature reaction and synchronous reactance / impedance
7. Equivalent circuit, load phasor diagrams and voltage regulation
8. Predetermination of regulation by EMF and MMF methods
9. Parallel operation of alternators and synchronising conditions
10. Dark-lamp, bright-lamp and synchroscope methods
11. Synchronising current, synchronising torque and synchronising power
12. Synchronous motor construction and principle of operation
13. Why the motor is not self-starting and practical starting methods
14. Equivalent circuit of the synchronous motor and effect of excitation
15. Original slide appendix for exact source-page reference

How this document is organised: the main body is written as clean study notes in the style of your Module 2 document. An appendix at the end embeds the original slide decks so that no source material is lost.

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Definition 1.1 — Synchronous Generator

A synchronous generator or alternator is an AC generator in which the rotor magnetic field is excited by DC and driven by a prime mover so that a 3-phase emf is induced in the stator armature windings.

1.1 Physical Idea

- The rotor carries the field winding; the stator carries the armature winding.
- DC excitation establishes fixed rotor poles.
- When the rotor is driven mechanically, the rotor field sweeps past the stationary stator conductors.
- Hence the flux linking each stator phase changes continuously and an emf is induced according to Faraday's law.
- Since the stator contains a 3-phase distributed winding, the induced voltages are 3-phase and displaced by 120° electrical.

1.2 Faraday-Law Basis

$$e = N\frac{d\Phi}{dt}$$

For a rotating coil in a sinusoidally distributed field, the flux linkage varies sinusoidally; hence the induced emf is also sinusoidal.

AC Generation in One Line

Mechanical rotation $\rightarrow$ changing magnetic flux linkage $\rightarrow$ induced alternating emf.

1.3 Why Slip Rings are Used

- If the output were taken directly from the rotating coil, the connecting wires would twist.
- Hence slip rings and brushes are used for a rotating coil arrangement.
- In a practical alternator, however, the armature is stationary and only the rotor field needs DC excitation through slip rings.

2.1 Classification

Basis	Types
Prime mover	Hydro generators, turbo generators, diesel-engine generators, gas-turbine generators
Number of phases	Single-phase, three-phase
Rotating member	Rotating armature, rotating field
Rotor construction	Salient-pole machine, non-salient (cylindrical / turbo) machine

2.2 Standard Construction

- Stator: houses the armature conductors in slots on the inner periphery.
- Rotor: carries the field poles and field winding supplied with DC.
- Excitation: supplied either from an external DC source through slip rings or from a shaft-mounted exciter.

Why Rotating Field and Stationary Armature are Preferred

- High voltage can be generated at the stationary armature terminals.
- Better insulation is possible for the armature winding.
- Collection of output current is easier.
- Rotor becomes lighter, especially for large machines.
- Only two slip rings are required for the field supply.

2.3 Salient-Pole vs Cylindrical Rotor

Feature	Salient-Pole Rotor	Cylindrical Rotor
Rotor form	Projecting poles mounted on rotor body	Smooth solid cylindrical rotor
Air gap	Non-uniform	Uniform
Shape	Large diameter, short axial length	Small diameter, long axial length
Speed range	Low / medium speed	High speed
Number of poles	Usually many poles	Usually 2 or 4 poles
Mechanical strength	Lower	Very high
Typical use	Hydroelectric stations	Steam / turbo alternators
Cooling	Natural / air cooling	Hydrogen or forced cooling possible

2.4 Constructional Notes from the Slides

- The armature core is built from laminated silicon-steel sheets to reduce hysteresis and eddy-current losses.
- Cylindrical rotors have low windage loss, good balancing and quieter operation.
- Salient-pole machines are well suited to low-speed prime movers such as water turbines.

3.1 Speed-Frequency Relation

$$f = \frac{PN_s}{120} \qquad \Longleftrightarrow \qquad N_s = \frac{120f}{P}$$

where f is electrical frequency in hertz, N_s is synchronous speed in rpm, and P is the number of poles.

The generator is called synchronous because the electrical frequency is locked to rotor speed and pole number.

3.2 Pitch Factor, Distribution Factor and Winding Factor

Definition 3.1 — Pitch Factor / Coil-Span Factor

The pitch factor k_p accounts for the reduction in generated emf when a coil is short-pitched.

$$k_p = \cos\frac{\alpha}{2}$$

where α is the chording angle.

Definition 3.2 — Distribution Factor

The distribution factor k_d accounts for the reduction in resultant emf when the phase winding is distributed over several slots.

$$k_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{m\sin\left(\frac{\beta}{2}\right)}$$

where m is the number of slots per pole per phase and β is the slot angle.

Winding Factor

$$k_w = k_p k_d$$

For sinusoidal flux distribution, k_w is used directly in the emf equation.

3.3 EMF Equation

$$E_{\text{ph}} = 4.44\, k_w\, f\, \Phi_m\, T$$

where T is the number of series turns per phase and Φ_m is flux per pole in weber.

$$E_{\text{ph}} = 4.44\, k_p k_d\, f\, \Phi_m\, T$$

Example — Generated EMF per Phase

Given: $f = 50\,\text{Hz}$, $\Phi_m = 0.05\,\text{Wb}$, $T = 100$ turns/phase, $k_p = 0.96$, $k_d = 0.95$. Find the generated emf per phase.

$$k_w = k_p k_d = 0.96 \times 0.95 = 0.912$$

$$E_{\text{ph}} = 4.44 \times 0.912 \times 50 \times 0.05 \times 100 \approx 1012 \text{ V}$$

If the winding is star connected, line voltage $\approx \sqrt{3}\, E_{\text{ph}} \approx 1753$ V.

Definition 4.1 — Open-Circuit Characteristic

The open-circuit characteristic (OCC) of an alternator is the graph between the no-load generated emf and field current when the machine runs at constant rated speed with the armature open-circuited.

4.1 Features of OCC

- At zero field current, a small residual emf is present because of residual magnetism.
- Initially, the OCC is almost linear; this region is called the air-gap line.
- At higher excitation, magnetic saturation sets in and the curve bends.

4.2 Why OCC is Important

- It gives the relation between field current and generated voltage.
- It is used for predetermination of voltage regulation.
- Together with short-circuit data, it helps determine synchronous impedance.

On no-load, the terminal voltage of the alternator is practically equal to the internally generated emf per phase because armature current is zero.

Definition 5.1 — Armature Reaction

Armature reaction is the effect of armature-flux mmf on the main field flux in the alternator.

5.1 Nature of Armature Reaction

Load Condition	Effect of Armature Reaction
Unity power factor	Cross-magnetising
Zero power factor lagging	Demagnetising
Zero power factor leading	Magnetising
General lagging power factor	Partly demagnetising and partly cross-magnetising
General leading power factor	Partly magnetising and partly cross-magnetising

5.2 Leakage Reactance and Synchronous Reactance

- Some flux produced by armature current does not cross the air-gap in the main path; it forms leakage flux.
- Leakage flux produces a voltage drop represented by the leakage reactance X_l.
- Armature reaction is often represented by an equivalent reactance X_a.

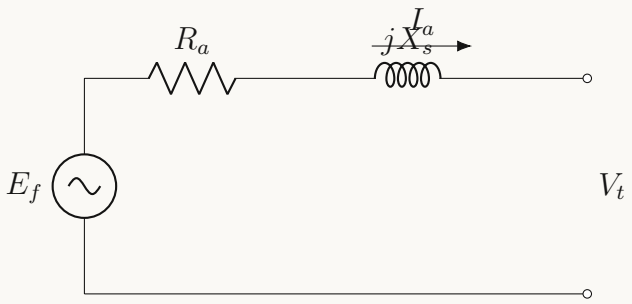
$$X_s = X_l + X_a \qquad \text{(synchronous reactance)}$$

$$Z_s = \sqrt{R_a^2 + X_s^2} \qquad \text{(synchronous impedance)}$$

Practical Meaning

The synchronous reactance combines the effect of armature leakage and armature reaction into one equivalent series reactance.

6.1 Per-Phase Equivalent Circuit of an Alternator



$$E_f = V_t + I_a(R_a + jX_s)$$

If R_a is neglected,

$$E_f \approx V_t + jI_aX_s$$

6.2 Phasor Behaviour with Different Loads

- Lagging load: armature current lags terminal voltage; terminal voltage tends to fall because the reactive drop adds strongly.
- Unity power factor load: armature current is nearly in phase with terminal voltage; armature reaction is mainly cross-magnetising.
- Leading load: armature reaction becomes magnetising; terminal voltage may rise with load and voltage regulation can become negative.

6.3 Load Characteristic

- The load characteristic is the relation between terminal voltage and armature current at constant speed and constant excitation.
- For lagging power-factor load, terminal voltage falls more rapidly.
- For leading power-factor load, voltage fall is smaller and may even reverse.

Definition 7.1 — Voltage Regulation

Voltage regulation of an alternator is the change in terminal voltage when the full-load is removed, while speed and excitation are kept constant, expressed as a fraction or percentage of full-load voltage.

$$\%\,\text{Voltage Regulation} = \frac{E_0 - V_t}{V_t} \times 100$$

where E_0 is the no-load emf corresponding to the same excitation and V_t is the full-load terminal voltage.

7.1 EMF Method (Synchronous Impedance Method)

Procedure

1. Determine the synchronous impedance Z_s from OCC and short-circuit data.
2. Calculate $X_s = \sqrt{Z_s^2 - R_a^2}$.
3. Use the phasor equation $E_f = V_t + I_a(R_a + jX_s)$.
4. Obtain $E_0 = |E_f|$ and then compute voltage regulation.

$$E_0 = \sqrt{(V_t \cos\phi + I_a R_a)^2 + (V_t \sin\phi \pm I_a X_s)^2}$$

$+$ for lagging power factor, $-$ for leading power factor.

The EMF method usually gives a higher value of regulation and is therefore called a pessimistic method.

7.2 MMF Method (Ampere-Turn Method)

Idea of the Method

- Convert the required terminal-voltage emf into an equivalent field current from OCC.
- Convert the armature reaction and leakage effects into an equivalent field current.
- Add these field-current components vectorially according to power factor.
- From the resultant field current, read the no-load emf from OCC and compute regulation.

The MMF method usually gives a lower value of regulation and is therefore regarded as an optimistic method.

8.1 Why Alternators are Operated in Parallel

- Better efficiency over varying load.
- Greater reliability and continuity of service.
- Maintenance and repair can be done without complete shutdown.
- Large power demand can be met by combining smaller units.

8.2 Requirements / Desirable Features

- Same output voltage rating.
- Speeds such that the machines generate the same frequency.
- Same phase sequence and same waveform.
- Similar speed-load characteristics of prime movers.
- Armature reactance present so current sharing is stable.

8.3 Conditions for Synchronising

Conditions to be Satisfied Before Closing the Switch

1. Incoming machine terminal voltage must equal bus-bar voltage.
2. Incoming machine frequency must equal bus-bar frequency.
3. Phase sequence must be identical.
4. Phase angle between corresponding voltages must be zero at the instant of switching.

Definition 8.1 — Infinite Bus-Bar

An infinite bus-bar is an electrical system of practically constant voltage and frequency, unaffected by the connection or disconnection of a single alternator.

8.4 Methods of Synchronisation

Method	Main Observation	Switch Closing Instant
Dark-lamp method	Lamps become dark when voltages match in phase	Middle of dark period
Bright-lamp method	Cross-connected lamps become bright at synchronism reference	Middle of bright period
Three-dark-lamp method	Three lamps brighten and darken together if phase sequence is correct	Instant all three are dark together
Two-bright-one-dark method	One lamp dark and two bright at synchronism	Lamp L_2 dark, other two bright
Synchroscope	Pointer shows relative phase and frequency difference	Pointer at vertical synchronising mark

8.5 Useful Observations from Lamp Methods

- If lamps brighten and dim in sequence, the phase sequence is wrong; interchange any two leads.
- In two-bright-one-dark method, a clockwise travelling light wave indicates one speed condition and the opposite direction indicates the other.
- The incoming machine is adjusted until lamp changes are very slow, then the switch is closed at the proper instant.

9.1 Synchronising Current

- If two equal emfs are exactly in phase opposition, no circulating current flows.
- If one induced emf falls back by a small electrical angle δ, a resultant emf appears in the local circuit.

For a small angular displacement δ (in radians),

$$E_R \approx E\delta$$

and the synchronising current is

$$I_{sy} = \frac{E_R}{Z_{cs}} \approx \frac{E\delta}{Z_{cs}}$$

If circuit resistance is much smaller than reactance,

$$I_{sy} \approx \frac{E\delta}{X_{cs}}$$

9.2 Role of Synchronising Current

- It acts as a generating current for one machine and a motoring current for the other.
- Hence it produces a synchronising torque that accelerates the retarded machine and decelerates the advanced machine.
- Any small departure from synchronism therefore produces a restoring tendency.

Synchronising Power

The power exchanged between two parallel alternators due to synchronising current is called synchronising power. It is the power-transfer mechanism that helps maintain steady parallel operation.

Definition 10.1 — Synchronous Motor

A synchronous motor is a 3-phase AC motor in which the rotor, excited by DC, runs at the synchronous speed of the stator rotating magnetic field.

10.1 Construction

Part	Details
Stator / armature	Laminated core, slots on inner periphery, 3-phase winding usually star connected, frame of cast iron or welded steel
Rotor / field system	Field poles with field winding supplied by DC, excitation through slip rings and brushes or shaft-mounted exciter

10.2 Working Principle

- The 3-phase stator supply produces a rotating magnetic field at synchronous speed.
- The DC-excited rotor behaves like a magnet with fixed polarity.
- If the rotor is already near synchronous speed in the same direction as the rotating field, opposite stator and rotor poles attract and lock magnetically.
- The rotor then continues to run at synchronous speed with zero steady-state slip.

10.3 Why the Motor is Not Self-Starting

Reason

- At standstill, the stator poles reverse rapidly every half cycle.
- Hence the rotor experiences torque alternately in clockwise and anticlockwise directions.
- Due to inertia, the average starting torque is zero.
- Therefore a synchronous motor is not self-starting unless a special starting arrangement is provided.

11.1 Starting by an External DC Motor

- The synchronous motor is mechanically coupled to a DC shunt or compound motor.
- The DC motor brings the rotor near synchronous speed.
- AC supply is applied to the stator and then DC excitation is applied to the rotor field.
- Once magnetic locking takes place, the auxiliary DC motor is disconnected.

11.2 Starting by an AC Motor (Pony Motor)

- A small induction motor is used as a pony motor.
- It brings the synchronous motor rotor close to synchronous speed.

- After that, the stator AC supply and rotor field excitation are applied for magnetic locking.

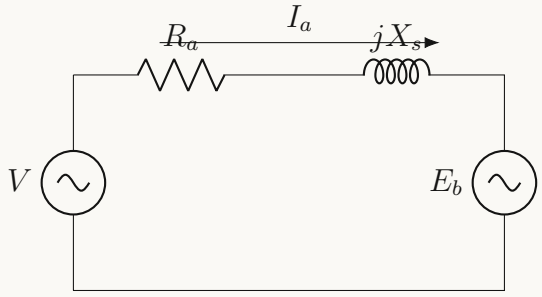
11.3 Starting by Damper Winding

- Copper bars embedded in the rotor pole faces and short-circuited at the ends form the damper winding.
- At start, the motor behaves like an induction motor.
- When it reaches near-synchronous speed, DC excitation is applied and the rotor pulls into step.

Most Practical Method

Damper winding is the most common practical method because it makes the machine effectively self-starting.

12.1 Equivalent Circuit of a Synchronous Motor



$$V = E_b + I_a(R_a + jX_s)$$

where E_b is the internal or back emf induced by the rotor field.

- The stator is supplied by 3-phase AC.
- The rotor is supplied by DC.
- Rotor flux induces a back emf E_b in the stator that opposes the applied voltage.

12.2 Effect of Excitation on Armature Current

Field Excitation	Armature Current	Power Factor
Under-excited	Higher than minimum	Lagging
Normal excitation	Minimum	Approximately unity
Over-excited	Increases again	Leading

V-Curve Idea

At constant load, a plot of armature current versus field current gives a V-curve. The current is minimum at the excitation corresponding to unity power factor.

When over-excited, a synchronous motor draws leading current and can be used for power-factor improvement as a synchronous condenser.

Aspect	Generator	Motor
Energy conversion	Mechanical to electrical	Electrical to mechanical
Internal emf relation	$E_f = V_t + I_a(R_a + jX_s)$	$V = E_b + I_a(R_a + jX_s)$
Rotor excitation	DC field required	DC field required
Speed	Locked to synchronous speed	Locked to synchronous speed
Starting nature	Prime mover starts it naturally	Not self-starting without auxiliary arrangement

Quantity	Formula / Key Relation
Synchronous speed	$N_s = \dfrac{120f}{P}$
Generated frequency	$f = \dfrac{PN_s}{120}$

(continued)

Quantity	Formula / Key Relation
Pitch factor	$k_p = \cos\dfrac{\alpha}{2}$
Distribution factor	$k_d = \dfrac{\sin(m\beta/2)}{m\sin(\beta/2)}$
Winding factor	$k_w = k_p k_d$
Alternator emf per phase	$E_{\text{ph}} = 4.44\, k_w\, f\, \Phi_m\, T$
Synchronous reactance	$X_s = X_l + X_a$
Synchronous impedance	$Z_s = \sqrt{R_a^2 + X_s^2}$
Generator phasor equation	$E_f = V_t + I_a(R_a + jX_s)$
Voltage regulation	$\%VR = \dfrac{E_0 - V_t}{V_t} \times 100$
EMF method	$E_0 = \sqrt{(V_t\cos\phi + I_aR_a)^2 + (V_t\sin\phi \pm I_aX_s)^2}$
Armature reaction at UPF	Cross-magnetising
Armature reaction at ZPF lagging	Demagnetising
Armature reaction at ZPF leading	Magnetising
Resultant emf for small angular displacement	$E_R \approx E\delta$
Synchronising current	$I_{sy} = \dfrac{E_R}{Z_{cs}} \approx \dfrac{E\delta}{Z_{cs}}$
Motor equivalent equation	$V = E_b + I_a(R_a + jX_s)$
Unity-pf excitation point	Minimum armature current on V-curve

1. In alternators, the field is on the rotor and the armature is on the stator.
2. Salient-pole rotors are for low-speed machines; cylindrical rotors are for high-speed turbo alternators.
3. Generated emf per phase depends on k_w, frequency, flux per pole and turns per phase.
4. At unity pf, armature reaction is mainly cross-magnetising; at lagging pf it is demagnetising; at leading pf it is magnetising.
5. EMF method is pessimistic; MMF method is optimistic.
6. Synchronising requires equality of voltage, frequency, phase sequence and phase angle.
7. A synchronous motor is not self-starting; damper winding solves this practically.
8. Over-excited synchronous motors draw leading current and help improve power factor.

Why the Appendix is Included

The notes above are rewritten for study. The original source slides are embedded below so that no diagram-only or slide-specific content is lost from the final single PDF.

PCEOT402
ELECTRICAL MACHINES

Module 3

Synchronous Machines

Three phase Synchronous Machine

- Principle of Operation of 3-phase synchronous generator – classification -
- constructional features - EMF equation – coil-span factor and distribution factor (sinusoidal flux distribution only) – numerical problems Synchronous generator on no-load – open circuit characteristics –
- synchronous generator on load – armature reaction – effect of armature reaction
- Equivalent circuit - phasor diagram – voltage regulation –
- predetermination of voltage regulation by emf and mmfmethod –
- Parallel operation - synchronous generator on infinite bus-bar – conditions – methods of synchronisation – dark lamp method – bright lamp method (2 hours)
- Synchronous motor – rotating magnetic field (no derivation) - principle of operation – starting methods

- Principle of Operation of 3-phase synchronous generator

AC Generator

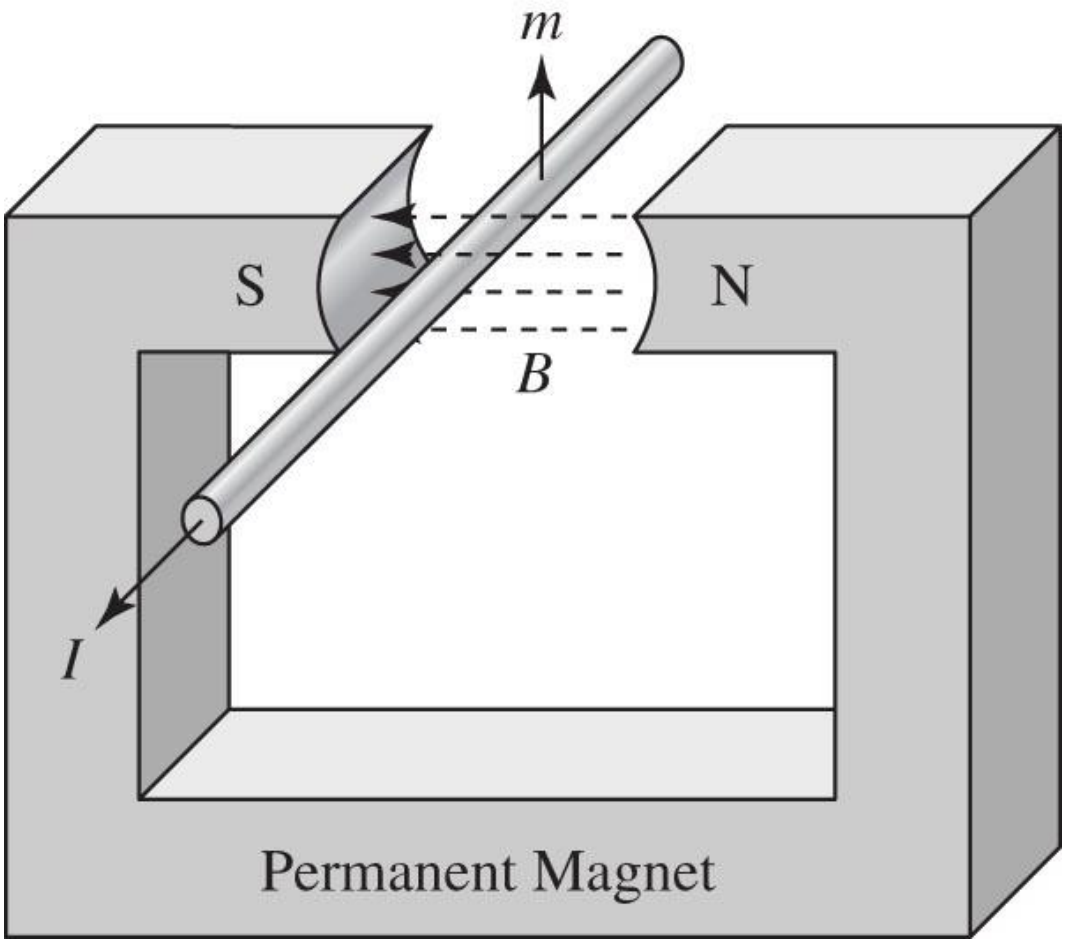
- Faraday's law dictates that if a coil of *N* turns experiences a change in magnetic flux, then the induced voltage *V* is given by

$$V = N\frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

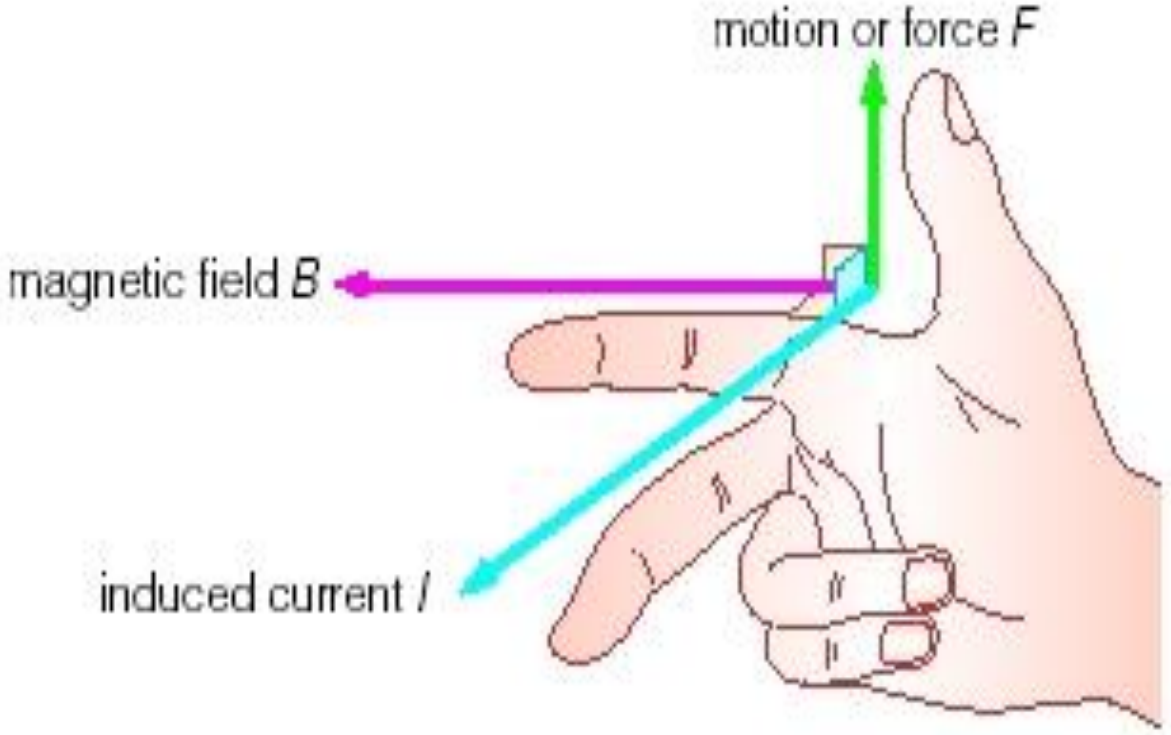
- If a coil of area *A* rotates with respect to a field *B*, and if at a particular time it is at an angle θ to the field, then the flux linking the coil is $BA\cos\theta$, and the rate of change of flux is given by

$$\frac{d\Phi}{dt} = BA\frac{\mathrm{d}(\sin\theta)}{\mathrm{d}t} = \frac{\mathrm{d}\theta}{\mathrm{d}t}\cos\theta = \omega\cos\theta$$

Magnetic induction in a wire moving in a field.



Fleming's Right hand rule

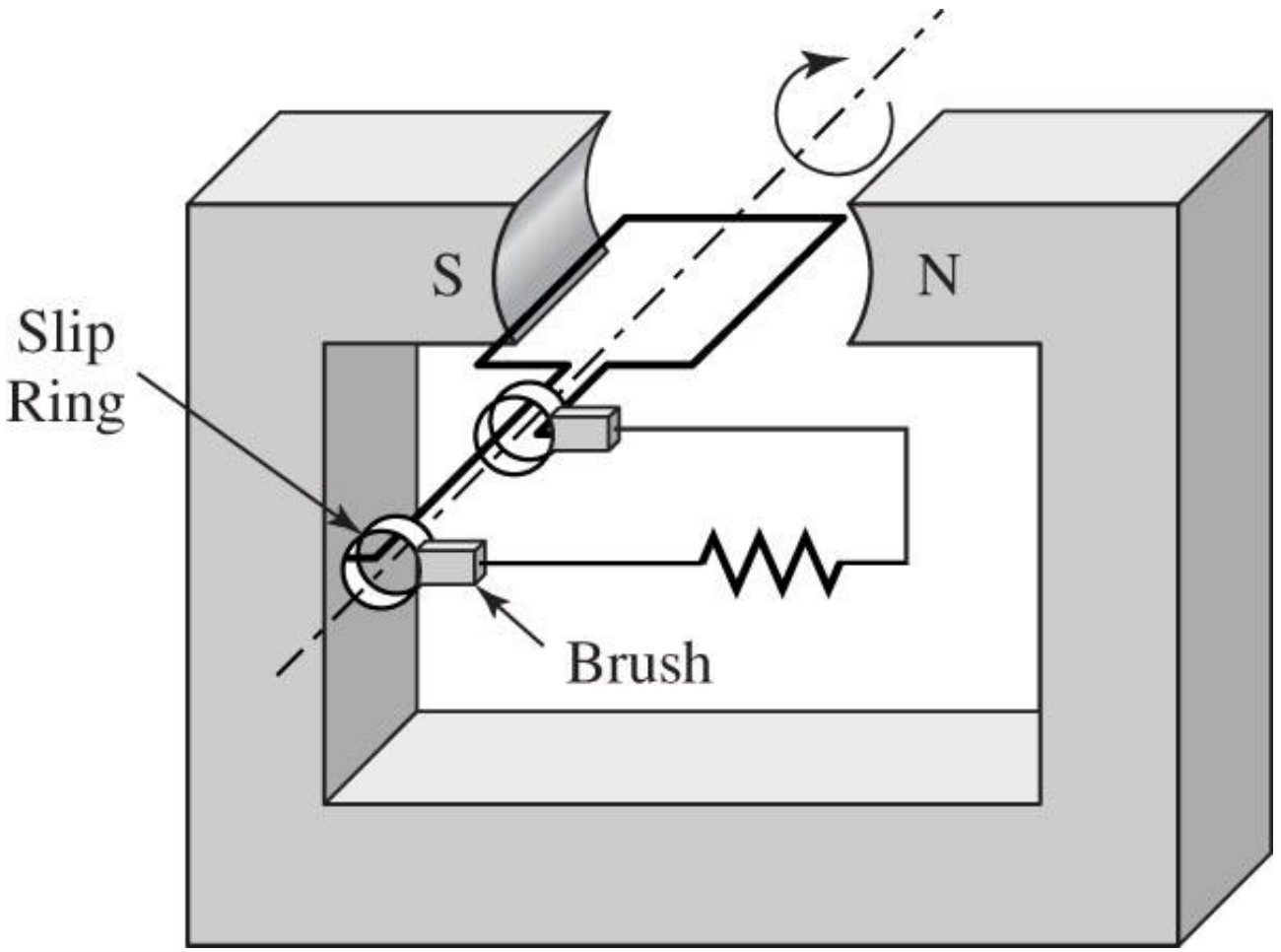


Fleming's Right hand rule

- Used to determine the direction of emf induced in a conductor
- The middle finger , the fore finger and thumb of the left hand are kept at right angles to one another.
  - The fore finger represent the direction of magnetic field
  - The thumb represent the direction of motion of the conductor
  - The middle finger will indicate the direction of the inducted emf .

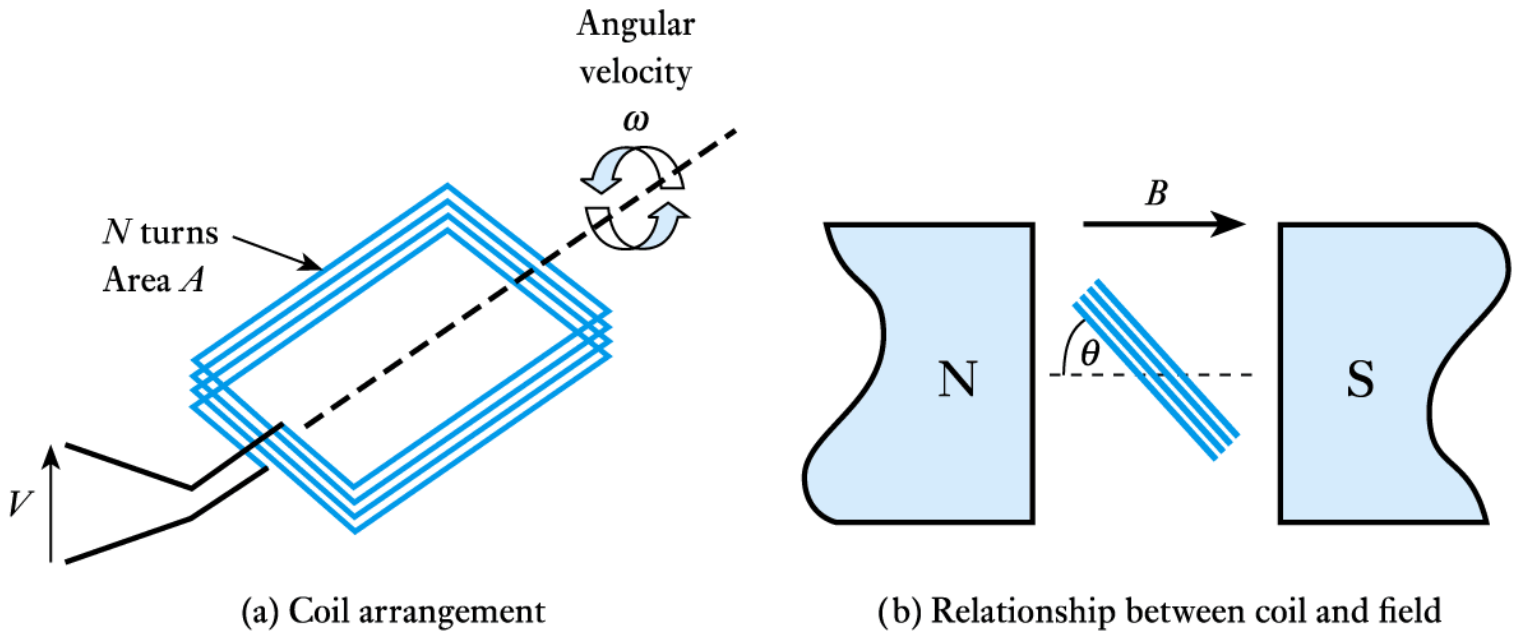
This rule is used in DC Generators

AC generator with slip rings and brushes.



- Thus for the arrangement shown below

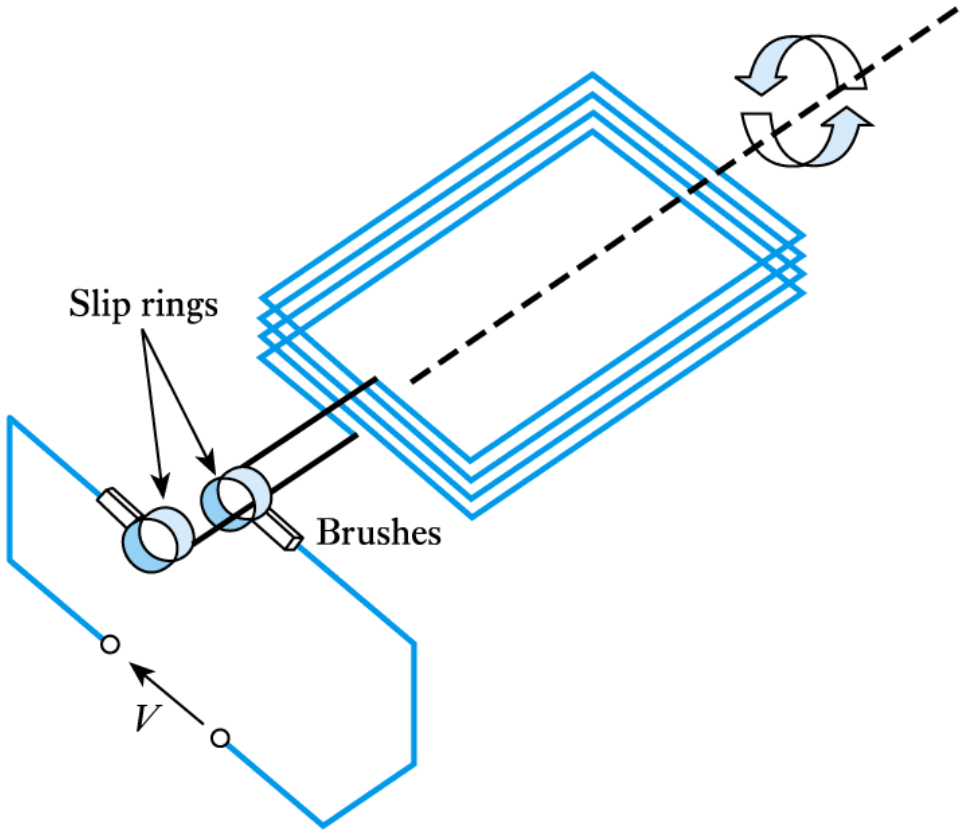
$$V = N\frac{\mathrm{d}\Phi}{\mathrm{d}t} = NBA\frac{\mathrm{d}(\sin\theta)}{\mathrm{d}t} = NBA\omega\cos\theta$$



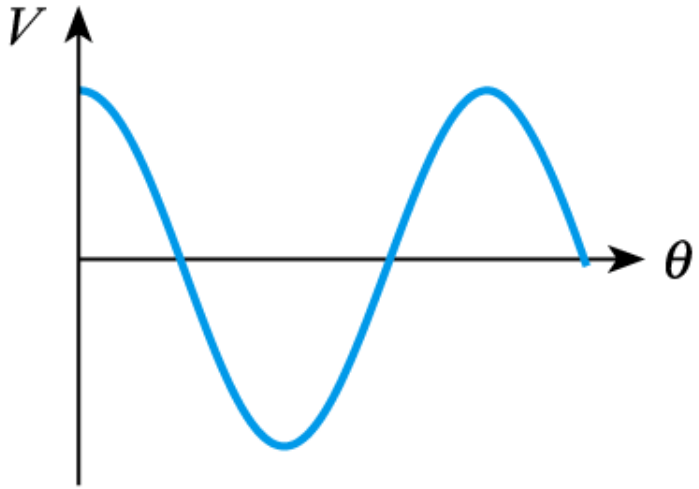
(a) Coil arrangement

(b) Relationship between coil and field

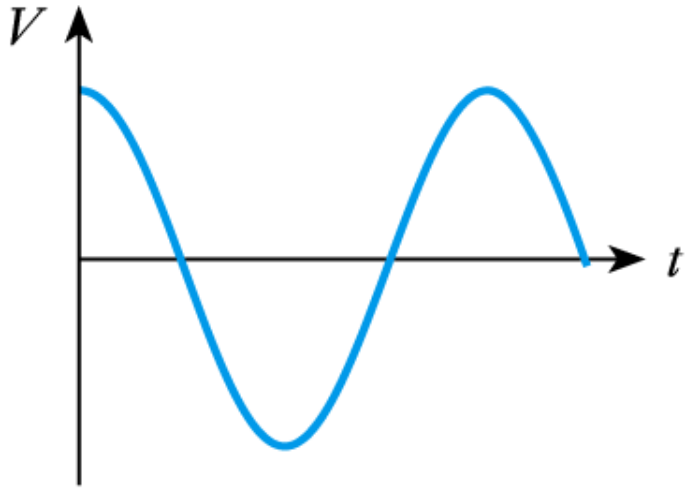
- Wires connected to the rotating coil would get twisted
- Therefore we use circular **slip rings** with sliding contacts called **brushes**



- Therefore this arrangement produces a sinusoidal output as shown below



(a)



(b)

SYNCHRONOUS GENERATOR CONSTRUCTION

- **rotor carry the "field windings" , while "armature windings" (or "stator windings") carry the main voltages of machine**

  in synchronous generator, rotor winding energized by dc source to develop rotor magnetic field

- rotor is turned by a prime mover, producing a rotating magnetic field which induce 3 phase voltages in stator windings

therefore:

- rotor windings - field windings
- stator windings - armature windings

3-phase synchronous generator – classification

Based on Prime Mover

- **Hydro Generators** – Driven by water turbines
- **Turbo Generators** – Driven by steam turbines
- **Diesel Engine Generators** – Driven by diesel engines
- **Gas Turbine Generators** – Driven by gas turbines

Based on No. of Phase

- Single phase
- Three phase

Based on type of roatating

- Rotating Armature
- Rotating Field

Based on Rotor Construction

- Salient Pole machine
- Non Salient Pole Machine

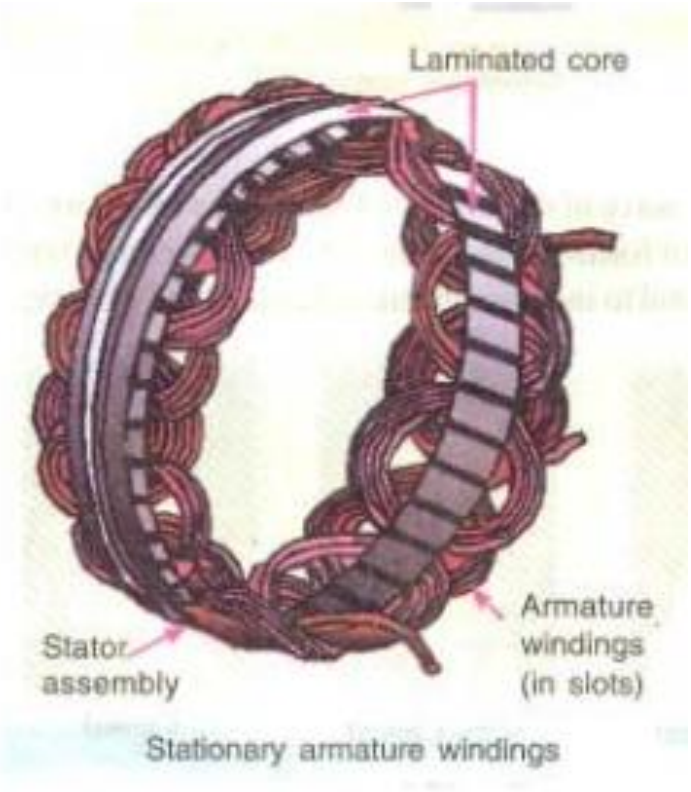
Advantages of rotating field and stationary armature type

- High voltage generation
- Better insulation
- Rotor weight – less
- Current collection - easy
- Lesser number of slip rings

Alternator - Construction

Standard construction consists of :

- Armature windings on stationery stator
- Field poles on rotor.
- The field windings are excited by DC supply from the exciter and given through the slip rings

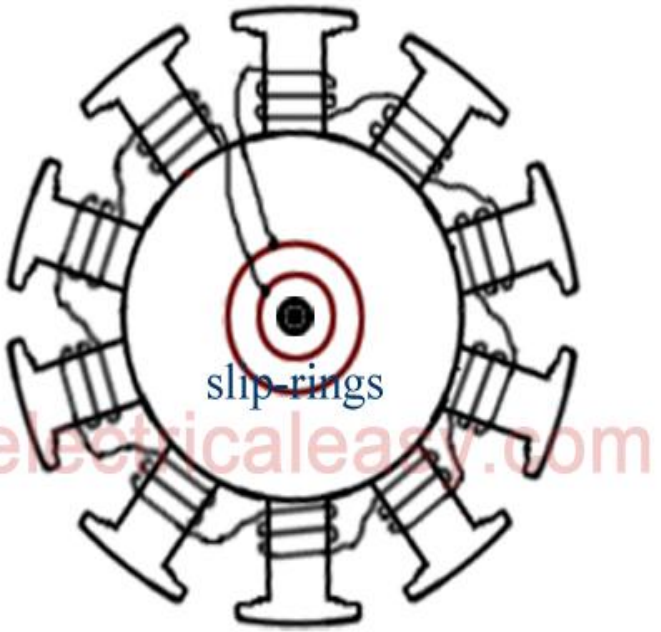


Types of alternators based on rotor construction

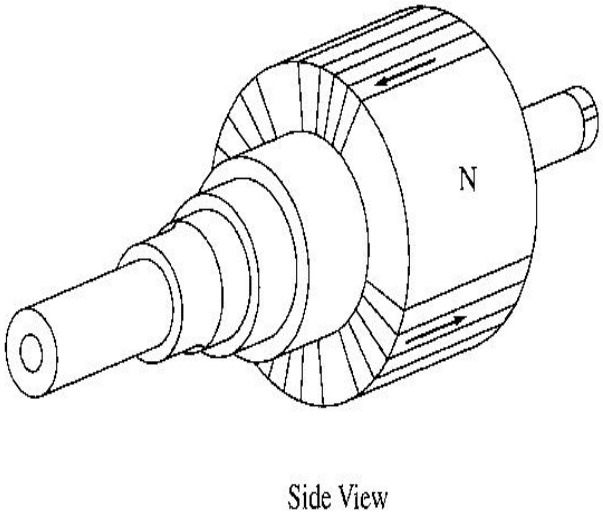
- Salient pole alternator
- Non salient pole or Turbo or Cylindrical rotor alternator

Rotor

Salient pole alternator

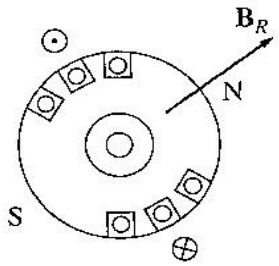


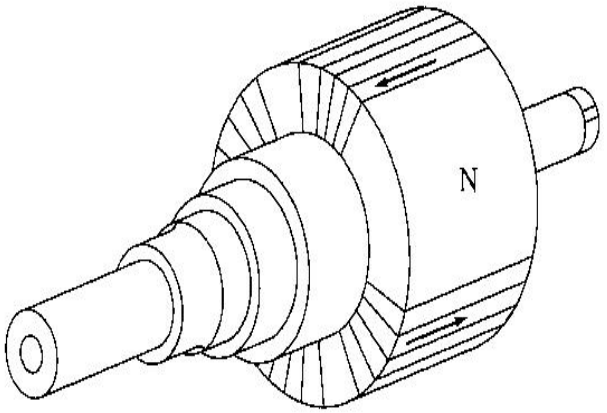
Non salient pole or Turbo or cylindrical rotor alternator



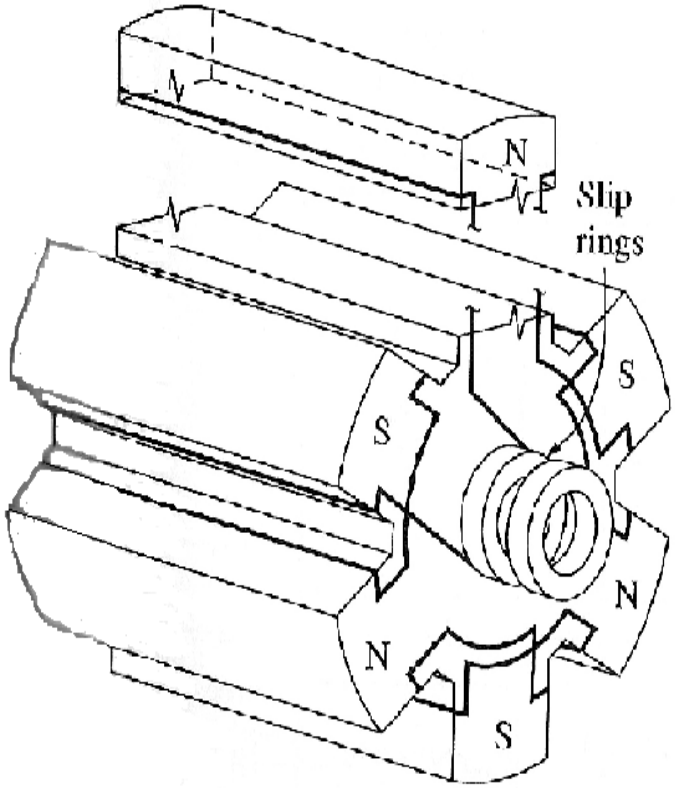
- Rotor of synchronous machine can be

Nonsalient: 2 pole rotor **Salient:** six-pole rotor





Side View



Salient pole type alternator

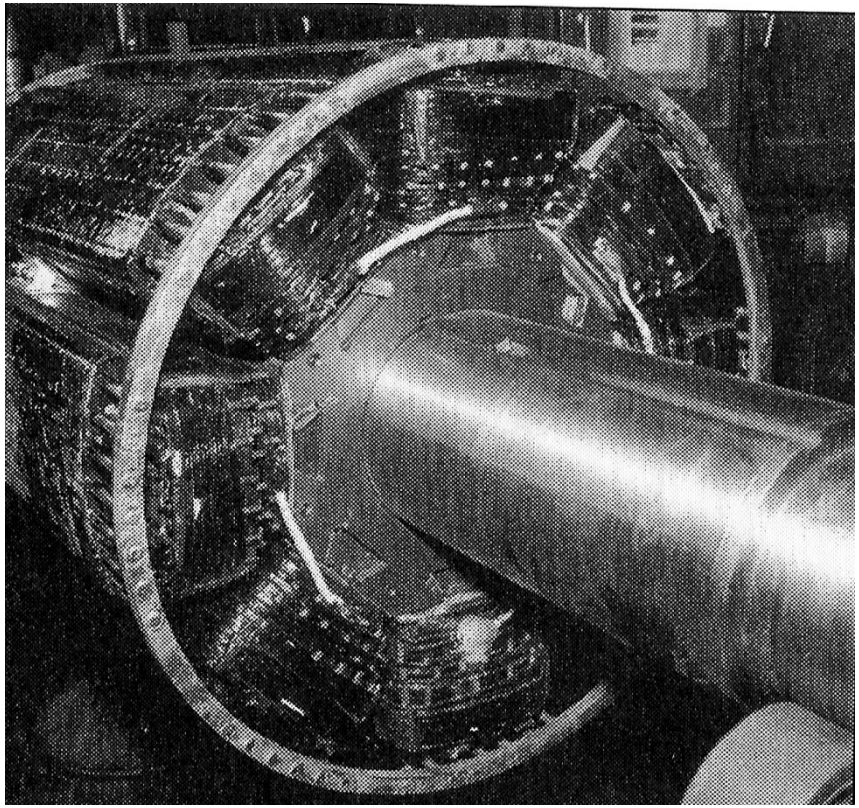
- It consists of cast iron frame which supports the silicon steel laminated (bundling of thin sheets) armature core .
- The core has slots on its inner periphery for housing the conductor .
- Silicon steel material reduces the hysteresis loss and the lamination structure reduces the eddy current loss.
- Suitable for low and medium speed alternator (Ex Hydro alternator)

Advantages of a smooth cylindrical type rotor

- Less windage loss
- No need of damper winding
- Good balancing of the rotor
- Sound less running
- Good cooling to the field because it is a distributed one.

SYNCHRONOUS GENERATOR CONSTRUCTION

- Photograph of a salient 8-pole synchronous machine rotor



SYNCHRONOUS GENERATOR CONSTRUCTION

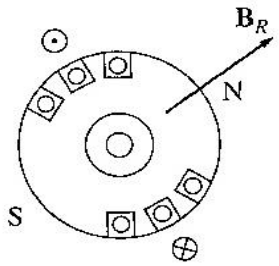
- **Rotor** experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
- To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply

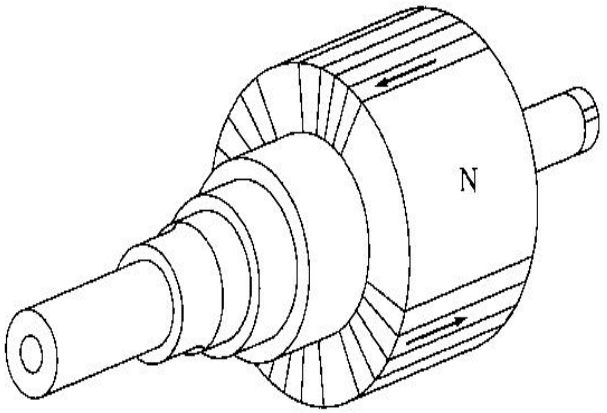
1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

SYNCHRONOUS GENERATOR CONSTRUCTION

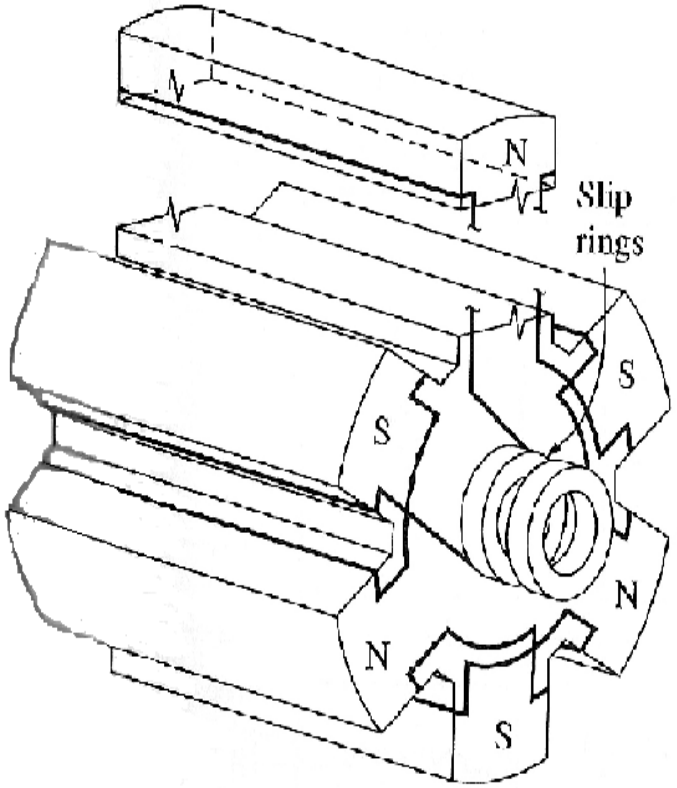
- Rotor of synchronous machine can be

Nonsalient: 2 pole rotor **Salient:** six-pole rotor





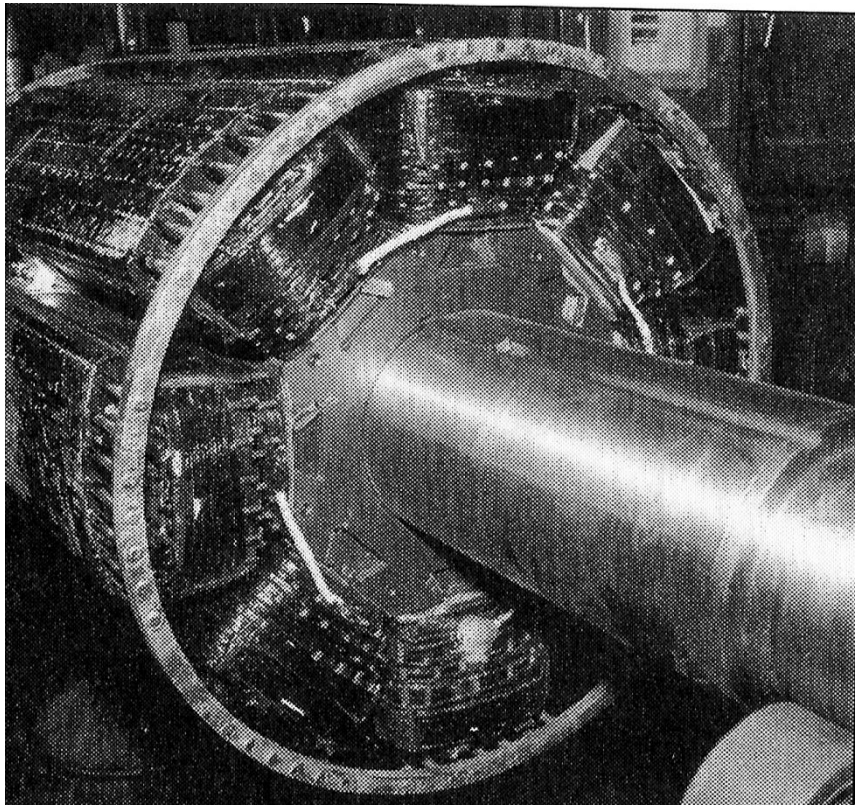
Side View



	Salient Pole Rotor	Non-Salient Pole (Cylindrical Rotor)
Rotor Construction	Projecting poles mounted on rotor	Smooth solid cylindrical rotor
Air Gap	Non-uniform air gap	Uniform air gap
Shape	Large diameter, short axial length	Small diameter, long axial length
Speed of Operation	Low speed (100–600 rpm)	High speed (1500–3000 rpm)
Number of Poles	Large number of poles (6–60 poles)	Usually 2 or 4 poles
Mechanical Strength	Lower mechanical strength	Very high mechanical strength
Applications	Hydroelectric power plants	Thermal & nuclear power plants
Cooling	Natural or air cooling	Hydrogen or forced cooling (in turbo generators)

SYNCHRONOUS GENERATOR CONSTRUCTION

- salient 8-pole synchronous machine rotor



SYNCHRONOUS GENERATOR CONSTRUCTION

- **Rotor** experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
- To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply

1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

paper.

Fig. 37.6

When the conductor is in the interpolar gap, as at *A* in Fig. 37.7, it has minimum e.m.f. induced in it, because flux density is minimum there. Again, when it is at the centre of a *S*-pole, it has maximum e.m.f. induced in it, because flux density at *B* is maximum. But the direction of the e.m.f. when conductor is over a *N*-pole is opposite to that when it is over a *S*-pole.

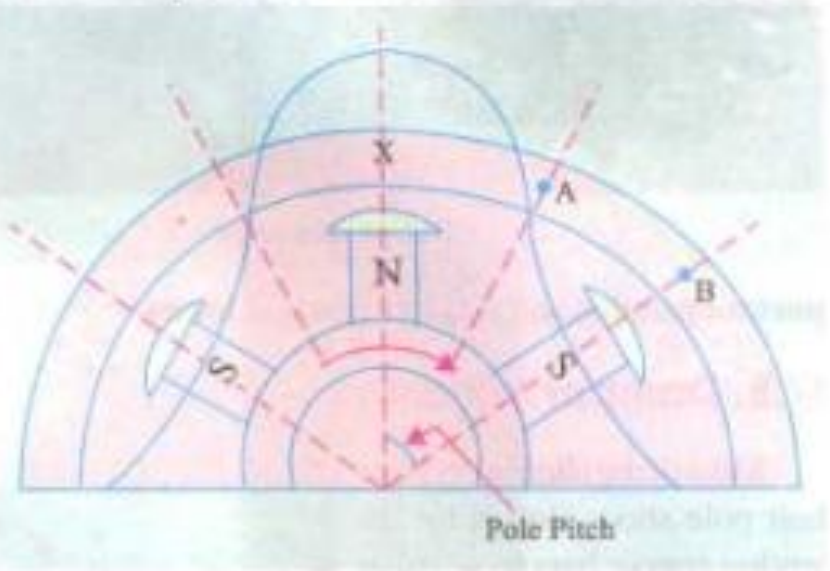


Fig. 37.7

Obviously, one cycle of e.m.f. is induced in a conductor when one pair of poles passes over it. In other words, the e.m.f. in an armature conductor goes through one cycle in angular distance equal to twice the pole-pitch, as shown in Fig. 37.7.

Let P = total number of magnetic poles

N = rotative speed of the rotor in r.p.m.

f = frequency of generated e.m.f. in Hz.

Since one cycle of e.m.f. is produced when a pair of poles passes past a conductor, the number of cycles of e.m.f. produced in one revolution of the rotor is equal to the number of pair of poles.

$\therefore$ No. of cycles/revolution = $P/2$ and No. of revolutions/second = $N/60$

$$\therefore \quad \text{frequency} = \frac{P}{2} \times \frac{N}{60} = \frac{PN}{120} \text{ Hz}$$

or
$$f = \frac{PN}{120} \text{ Hz}$$

N is known as the synchronous speed, because it is the speed at which an alternator must run, in

SYNCHRONOUS GENERATOR

Speed of rotation of synchronous generator

- synchronous generators are *synchronous, during their operation*

means: electrical frequency is synchronized with mechanical speed of rotor

- Relation between electrical frequency of stator and mechanical speed of rotor as shown before:
- $f = Ns\ p / 120$

f : electrical frequency in Hz

Ns: speed of rotor in r/min

p: number of poles

Winding factor

The product of pitch factor K_p and distribution factor K_d is referred to the winding factor

$$i.e.\ K_w = K_p K_d.$$

The pitch factor and distri...

Pitch factor

3. 6.1. Pitch Factor. In a short pitch winding the induced emfs in the two sides of the coil are not in phase, as discussed in Art 3.5, hence their resultant given by the phasor sum is always less than their arithmetic sum. The ratio of phasor sum of induced emfs per coil to the arithmetic sum of induced emfs per coil is known as *coil pitch factor, coil-span factor* or simply *pitch factor* (K_c or K_p). It is always less than unity.

Let the coil have a pitch short by an angle α electrical space degrees from full pitch and induced emf in each coil side be E.

Now if the coil would have been full-pitched, then total induced emf in the coil would have been 2 E.

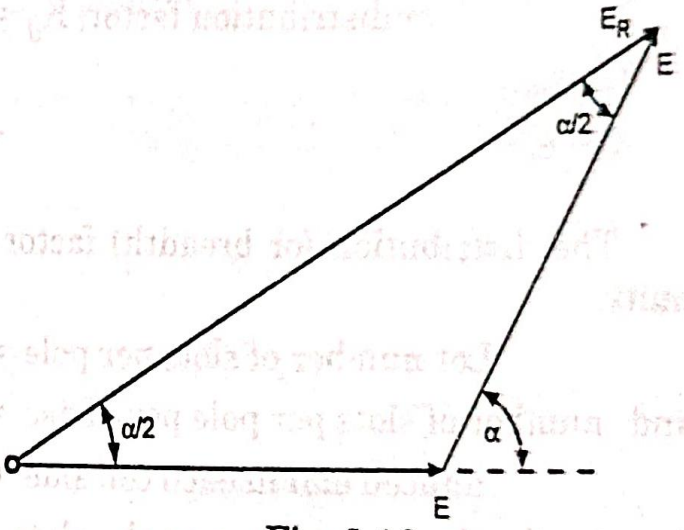


Fig. 3.16

If the coil is short-pitched by α electrical space degrees the resultant induced emf, E_R in the coil is phasor sum of two voltages α° (electrical) apart

$$i.e.\ E_R = 2\,E\cos\frac{\alpha}{2}$$

$$\text{Pitch factor, } K_p = \frac{\text{Phasor sum of coil side emfs}}{\text{Arithmetic sum of coil side emfs}} = \frac{2\,E\cos\frac{\alpha}{2}}{2\,E} = \cos\frac{\alpha}{2} \qquad ...(3.3)$$

Distribution factor

The ratio of the phasor sum of the emfs induced in all the coils distributed in a number of slots under one pole to the arithmetic sum of the emfs induced (or to the resultant of emfs induced in all coils concentrated in one slot under one pole) is known as breadth factor (K_b) or distribution factor (K_d)

$$\text{or distribution factor, } K_d = \frac{\text{EMF induced in distributed winding}}{\text{EMF induced if the winding would have been concentrated}}$$

$$= \frac{\text{Phasor sum of component emfs}}{\text{Arithmetic sum of component emfs}}$$

The distribution (or breadth) factor is always less than unity.

Let number of slots per pole $= n$

and number of slots per pole per phase $= m$

Induced emf in each coil side $= E_c$

Angular displacement between the slots,

$$\beta = \frac{180^\circ}{n}$$

The emfs induced in different coils of one phase under one pole are represented by side AC, CD, DE, EF, ...which are equal in magnitude (say each equal to E) and differ in phase (say by β°) from each other.

If bisectors are drawn on AC, CD, DE, EF... they would meet at common point (say at O). The point O would be the circum centre of the circle having AC, CD, DE, EF...as the chords and representing the emfs induced in the coils in different slots.

$$\text{EMF induced in each coil side, } E = AC = 2\,OA \sin\frac{\beta}{2}$$

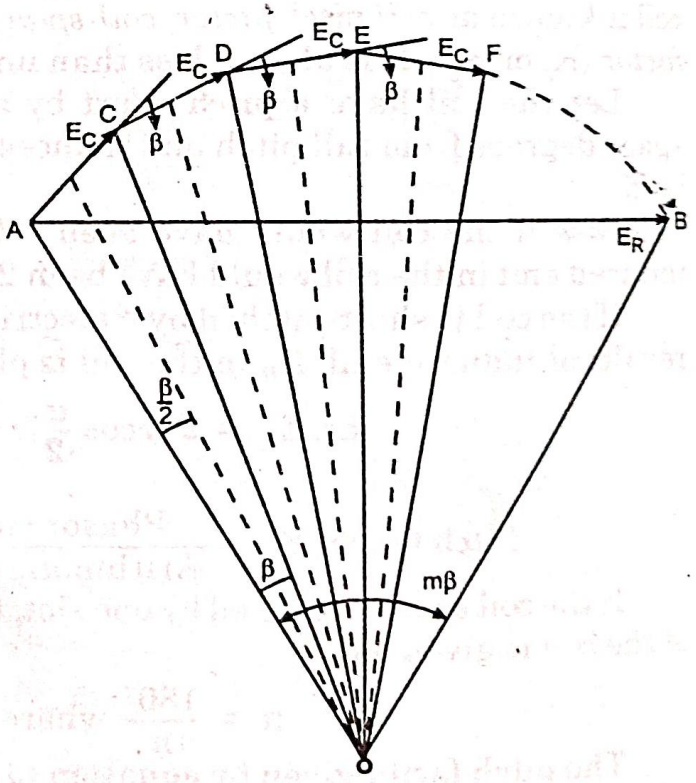


Fig. 3.17

$$\text{Arithmetic sum} = m \times 2 \times \text{OA} \sin \frac{\beta}{2}$$

The resultant emf induced in one polar group of one phase would be the phasor sum as represented by the phasor AB shown in fig 3.17.

$$\therefore \text{The resultant emf, } E_R = \text{AB} = 2 \times \text{OA} \sin \frac{\text{AOB}}{2} = 2 \times \text{OA} \sin \frac{m\beta}{2}$$

$$\text{and distribution factor, } K_d = \frac{\text{Phasor sum of components emfs}}{\text{Arithmetic sum of components emfs}}$$

$$= \frac{2 \times \text{OA} \sin \frac{m\beta}{2}}{m \times 2 \times \text{OA} \sin \frac{\beta}{2}} = \frac{\sin \frac{m\beta}{2}}{m \sin \frac{\beta}{2}} \qquad \ldots(3.5)$$

$m\beta$ is also known as the *phase spread* and is spread in electrical degrees or radians.

The distribution factor K_d given by equation (3.5) is for the fundamental component of emf. If the flux distribution contains space-harmonics, the slot angular pitch β on the fundamental scale, would become $r\beta$ for the *r th* harmonic component and thus the distribution factor for the *r th* harmonic would be

$$K_{dr} = \frac{\sin \frac{rm\beta}{2}}{m \sin \frac{r\beta}{2}} \qquad \ldots(3.6)$$

EMF Equation

Let Z_p = Number of conductors in series per phase

P = Number of poles

ϕ = Useful flux per pole in webers

N = Rotational speed in rpm

and f = Frequency in herz and is equal to $\frac{NP}{120}$

The flux cut by any conductor while passing from the centre of one inter polar gap to the centre of the next is ϕ webers and since during the movement, the emf wave completes half cycle *i.e.* the time taken is $\frac{1}{2f}$ seconds, therefore, the average rate of cutting the flux,

$$\frac{d\phi}{dt} = \frac{\phi}{1/2f} = 2f\phi \text{ Wb/s}$$

Hence average emf induced in each conductor = $2f\phi$ volts

Average emf per phase, E_{av}/phase = Number of conductors in series/phase arranged in one slot/pole × average emf induced per conductor

$= Z_p \times 2f\phi = 2T \times 2f\phi = 4\phi f T$ volts $\quad \because Z_p = 2T$ (turns)

For distributed winding the average value of emf per phase will be K_d times the above value

i.e. E_{av} / phase $= 4K_d \phi f T$ volts

For short pitch winding the true average value of emf per phase will be K_p time above value

E_{av}/phase $= 4K_d K_p \phi f T$ volts

and E_{rms}/phase = Form factor, $K_f \times E_{av}$/phase $= 4K_f K_d K_p \phi f T$ volts

For sinusoidal wave of emf, $K_f = 1.11$

$\therefore E_{rms}$ / phase $= 4.44 K_d K_p \phi f T$ volts ...(3.7)

If the alternator is star-connected, as is usually the case, the line voltage is $\sqrt{3}$ times the phase value.

So line induced emf, $E_L = \sqrt{3} \times 4.44 K_d K_p \phi f T$ volts ...(3.8)

For full-pitched and concentrated windings, $K_p = 1$ and $K_d = 1$

PCEOT402
ELECTRICAL MACHINES

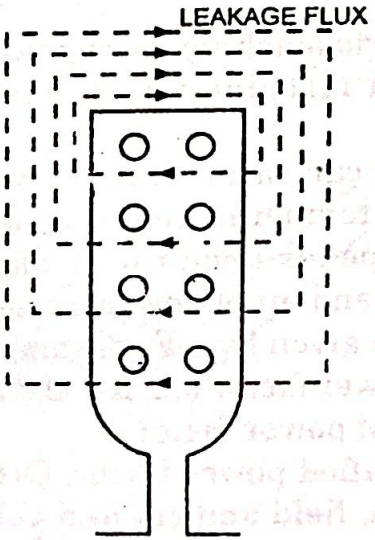
Module 3

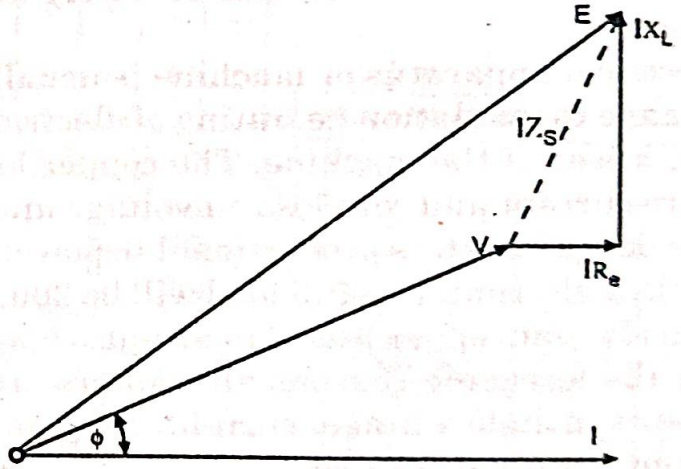
Synchronous Machines

Three phase Synchronous Machine

- Synchronous generator on no-load – open circuit characteristics –
- synchronous generator on load – armature reaction – effect of armature reaction
- Equivalent circuit - phasor diagram – voltage regulation –
- predetermination of voltage regulation by emf and mmf method
- Parallel operation - synchronous generator on infinite bus-bar –
- conditions – methods of synchronisation – dark lamp method – bright lamp method

Leakage reactance

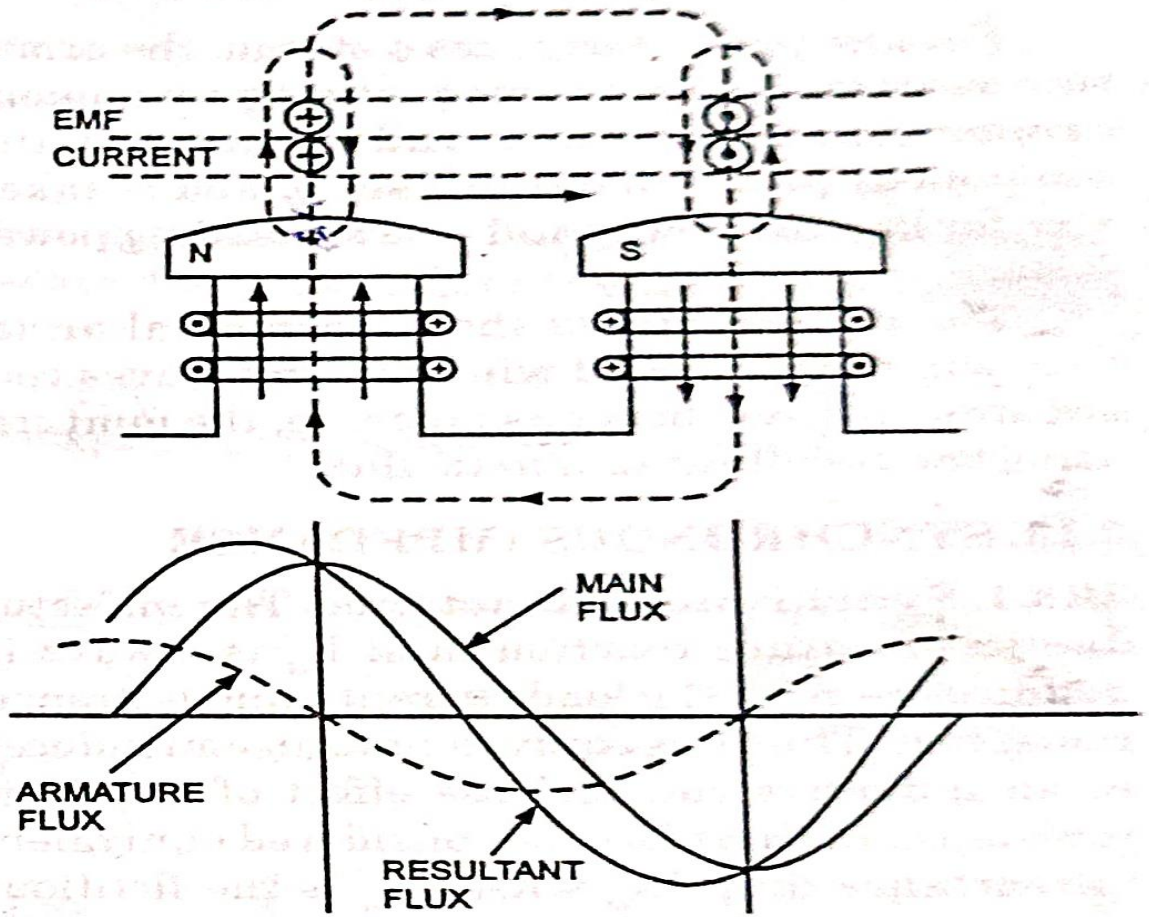




Armature Reaction

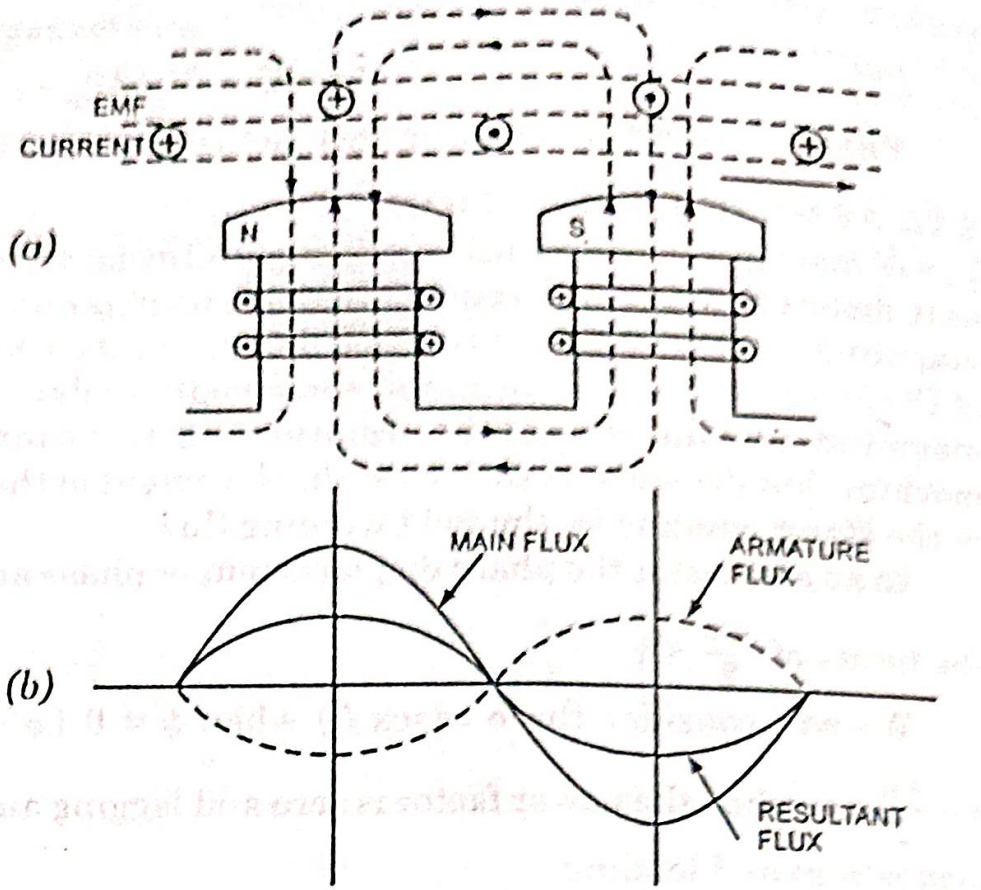
- Effect of armature flux on main flux
- In alternator it depend on type of load
- Effects are
    - Magnetising
    - Demagnetising
    - Cross magnetising

i) At upf load



Hence armature reaction at unity power factor has got distorting effect.

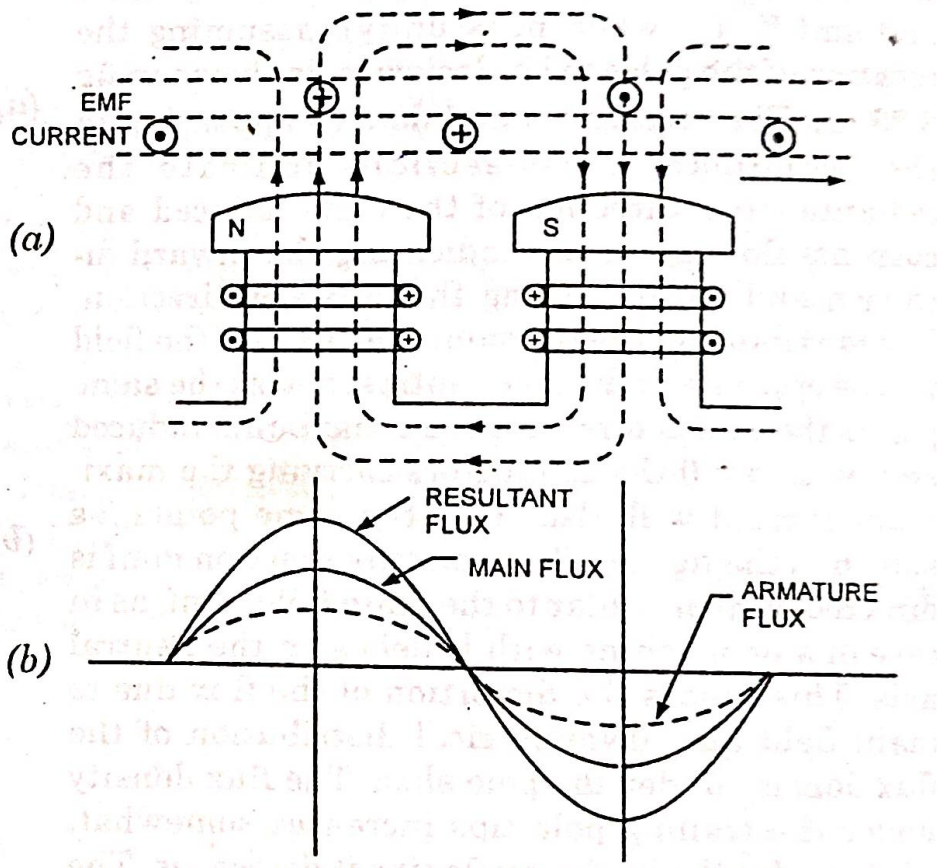
ii) Zero power factor lagging



Armature Reaction At Zero Power Factor (Lagging)

demagnetising effect.

iii) Zero power factor leading



Armature Reaction At Zero Power Factor (Leading)

Synchronous Impedance

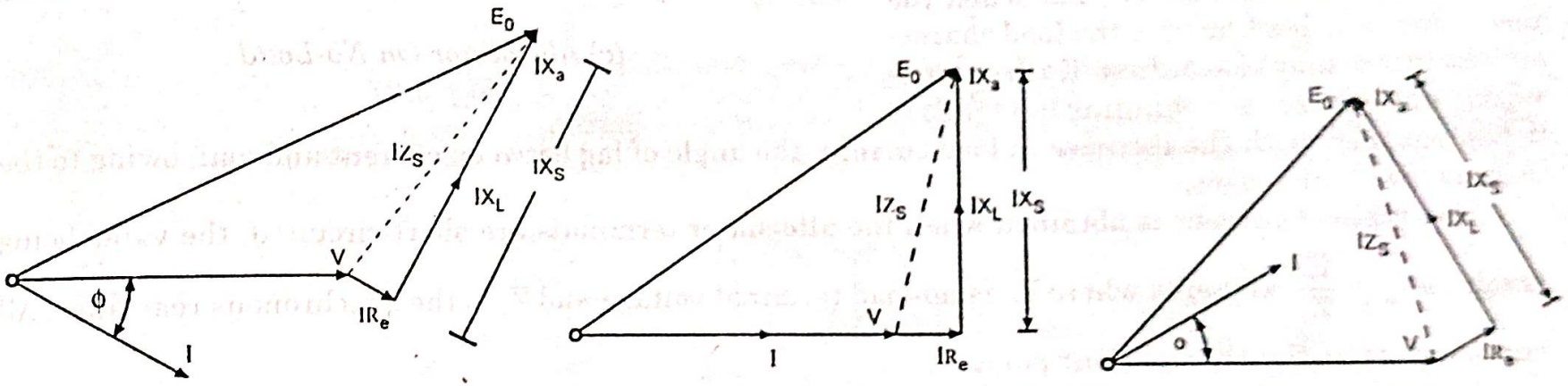
- Synchronous Reactance

$$X_s = X_L + X_a$$

- Synchronous Impedance

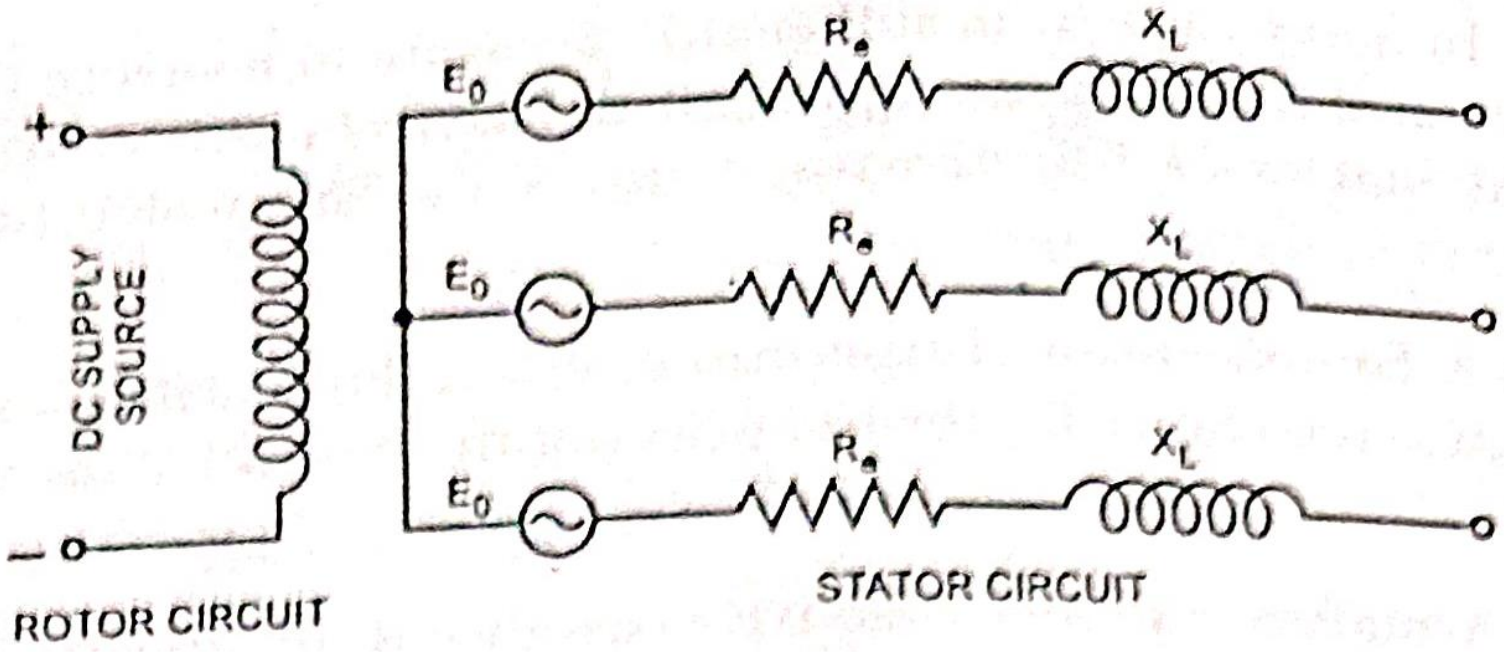
$$Z_s = R_a + jX_s$$

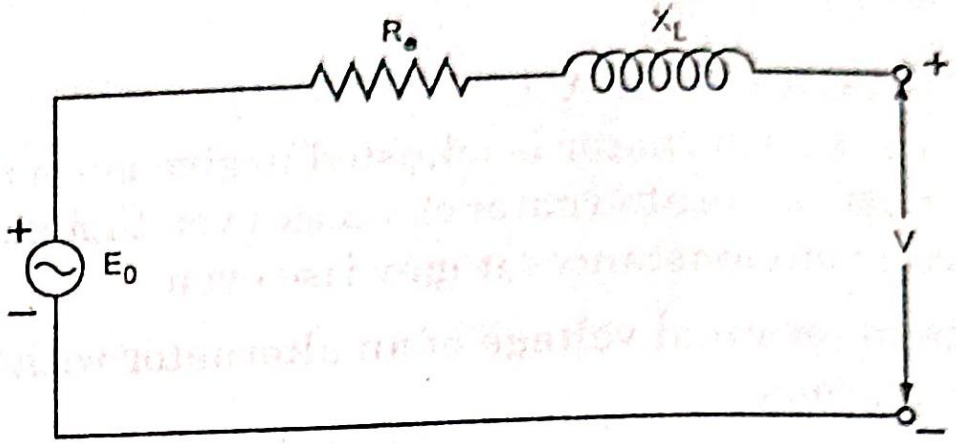
Alternator on load Phasor Diagram



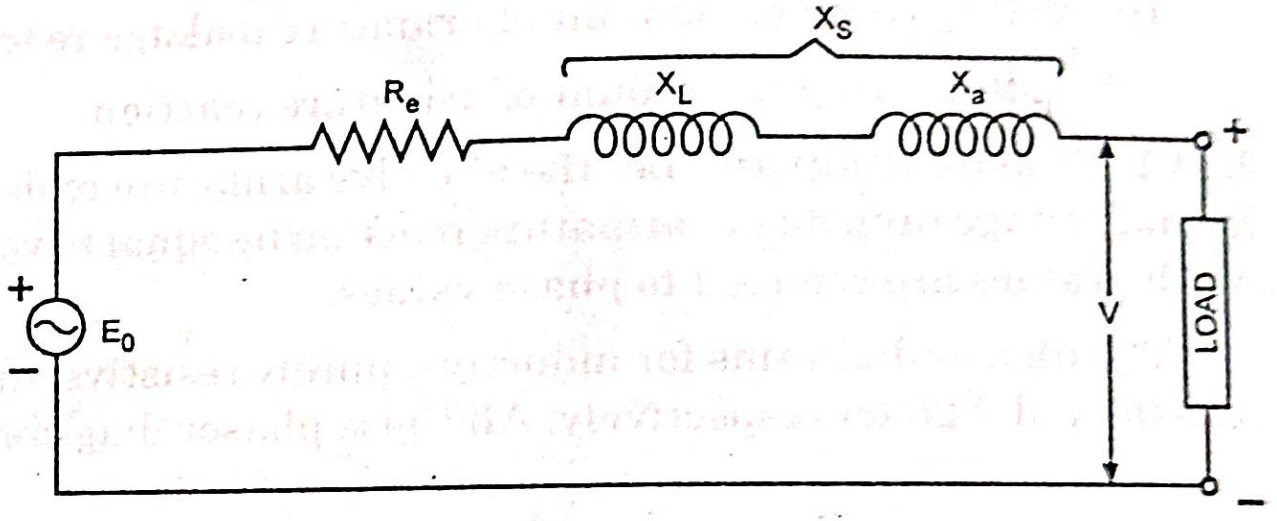
(a) At Lagging PF *(b) At Unity PF* *(c) At Leading PF*

Equivalent Circuit



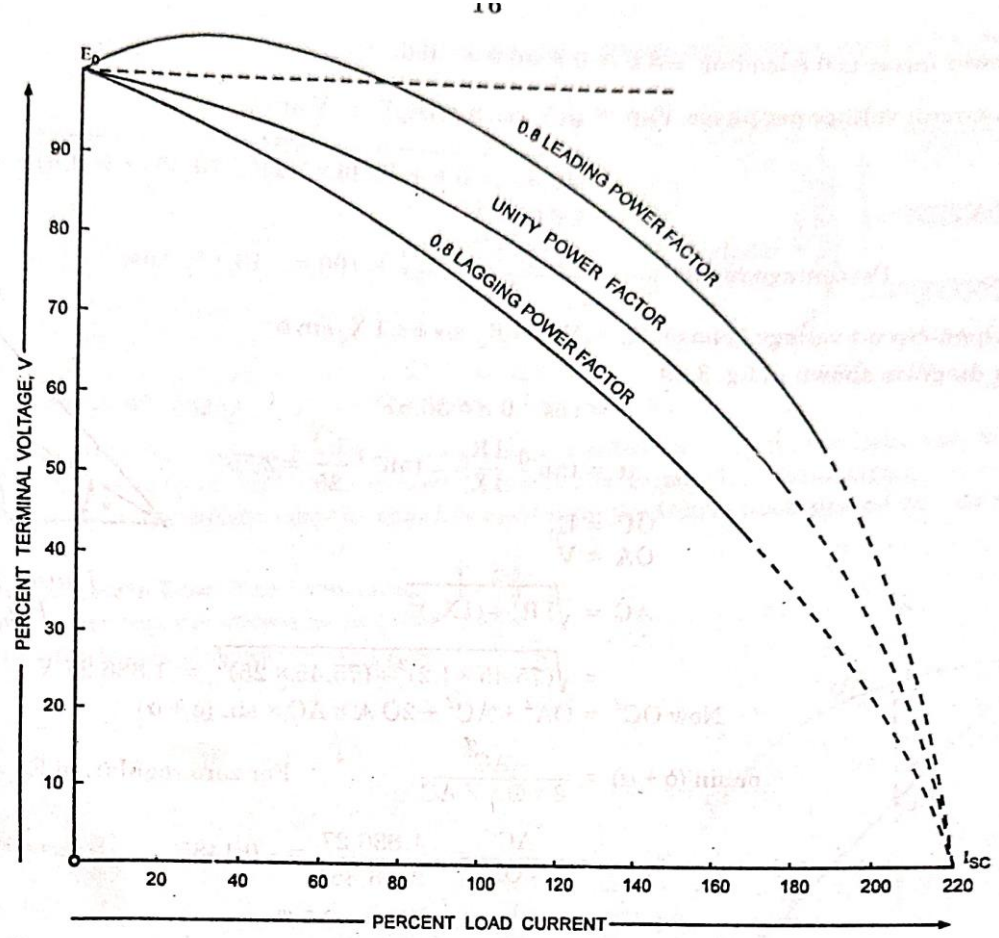


(b) Equivalent Circuit Diagram



(c) Alternator On No-Load

Load Characteristics



Voltage Regulation

i.e. voltage regulation $= \dfrac{E_0 - V}{V}$ per unit

$= \dfrac{E_0 - V}{V} \times 100$ in percentage

EMF Method (Synchronous Impedance Method)

nd short-circuit characteristic.

From short-circuit and open-circuit tests, the voltage egulation can be determined by any one of the two methods iven below.

1. Synchronous Impedance Method or EMF lethod. After determining R_a and X_s from open-circuit and ort-circuit tests, the phasor diagram for any load of any wer factor can be drawn, as shown in fig 3.35.

Let load current be I lagging behind the terminal phase ltage V by ϕ

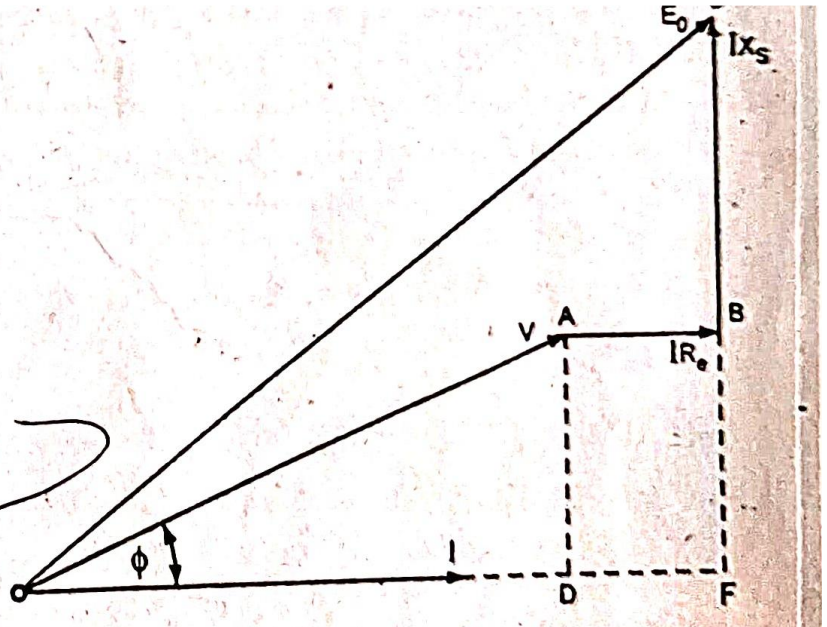


Fig. 3.35

No load terminal voltage per phase, E_0 = OC

Fulll-load terminal voltage, V = OA

Armature resistance drop, $I R_a$ = AB

Synchronous reactance drop $I X_s$ = BC

Now $OC^2 = OF^2 + FC^2 = (OD + DF)^2 + (FB + BC)^2$

or $E_0^2 = (V \cos \phi + I R_a)^2 + (V \sin \phi + I X_s)^2$

[Since BF = AD = V sin ϕ]

$$\text{or } E_0 = \sqrt{(V\cos\phi + IR_e)^2 + (V\sin\phi + IX_s)^2}$$

$$\text{And \% regulation} = \frac{E_0 - V}{V} \times 100$$

Note 1. For leading pf taking ϕ as negative and for unity pf taking $\phi = 0$ voltage regulation determined at leading pf and unity pf respectively.

2. Of various methods for determination of regulation, this one yields results most likely to be The regulation obtained by this method is always higher than actual value and, therefore, this is ca *pessimistic* method. The results are, however, more likely to be 'on safe side'. However, this m theoretically accurate for non-salient pole machines with distributed field windings when saturati considered. The reason of error is that in this method synchronous impedance or reactance is ass remain constant while actually it is not. The value of synchronous impedance (or reactance) varies saturation. At low saturation its value is larger because the effect of armature reaction is greater th at high saturation. Now under short-circuit conditions, saturation is very low and, therefore, the synchronous impedance measured is higher than that in actual working conditions

MMF (Ampere Turn Method)

synchronous impedance measured is higher than that in actual working conditions.

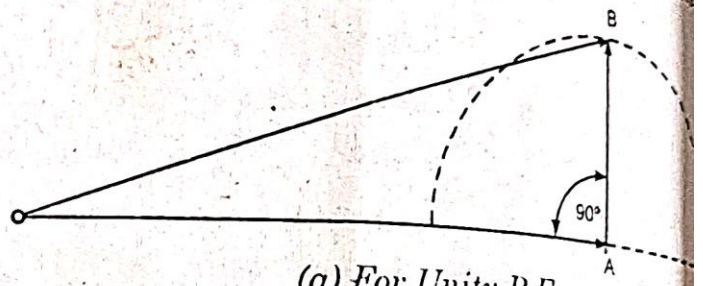
2. *Ampere-Turn (or MMF) Method.* In this method, the datas determined from open-circuit te
utilized.

From the above two tests the open-circuit and short-circuit characteristics are drawn, as shown in fig 3.32.

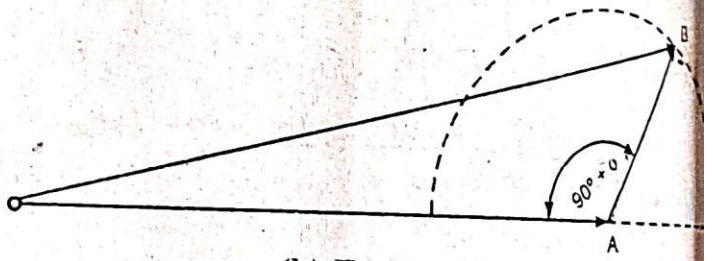
From the above characteristic field current I_{f1} is determined to give rated voltage V on no load, neglecting armature resistance drop, and field current I_{f2} is determined to cause short-circuit current, equal to full load current, on short circuit.

On short circuit, the field excitation I_{f2} balances the impedance drop in addition to armature reaction on full load. But since R_e is usually very small and X_L is also small for low voltage on short circuit, so impedance drop can be neglected. Hence pf on short circuit is almost zero lagging and the field amp-turns are used entirely to overcome the armature reaction. Therefore, I_{f2} gives demagnetising amp-turns at full load.

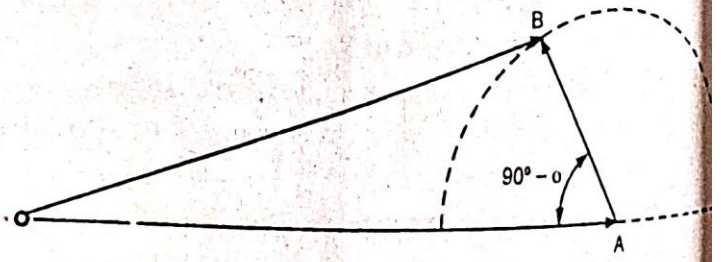
Now let the alternator supply full load current at a pf of cos ϕ. Draw OA representing I_{f1} to give full load rated voltage, V [or more exactly V + $I R_e \cos \phi$] then draw AB at an angle ($90^\circ \pm \phi$) representing I_{f2} to give full load current on short circuit; + ve sign for lagging pf and –ve sign for leading pf. Now find field current I_f measuring OB, which will give open-circuit EMF E_0, which can be determined from OCC.



(a) For Unity P F



(b) For Lagging P F



(c) For Leading P F

Parallel Operation of Alternators

- Efficiency
- Reliability or Continuity of Service
- Maintenance and Repair
- Physical Size

Requirements for Parallel Operation

- ✓ Alternators must have same output voltage rating
- ✓ Speeds of the machines should be such as to generate same frequency
- ✓ Alternators should be of same type so as to generate voltages of same waveform
- ✓ Prime movers of the alternators should have same speed-load characteristics
- ✓ Alternators should have reactances in their armature

Conditions for Synchronising

* **Synchronising** - The process of connecting an alternator in parallel with another alternator or with the common bus bar.

* For satisfactory operation,

- The terminal voltage of incoming machine must be exactly equal to that of running machines or common bus bar
- The speed of incoming machine must be such that its frequency *f*

equals to bus bar frequency

- Phase of incoming machine voltage must be the same as that of the bus bar voltage relative to the load.
- For three phase alternator, the phase sequence of incoming machine must be same as that of bus bar.

Methods of Synchronisation

- Synchronising of three phase alternators
  - Three Dark Lamp Method
  - Bright lamp Method (Two Bright and One Dark Lamp Method)
  - Synchroscope

Synchronization of 1-Phase Alternators

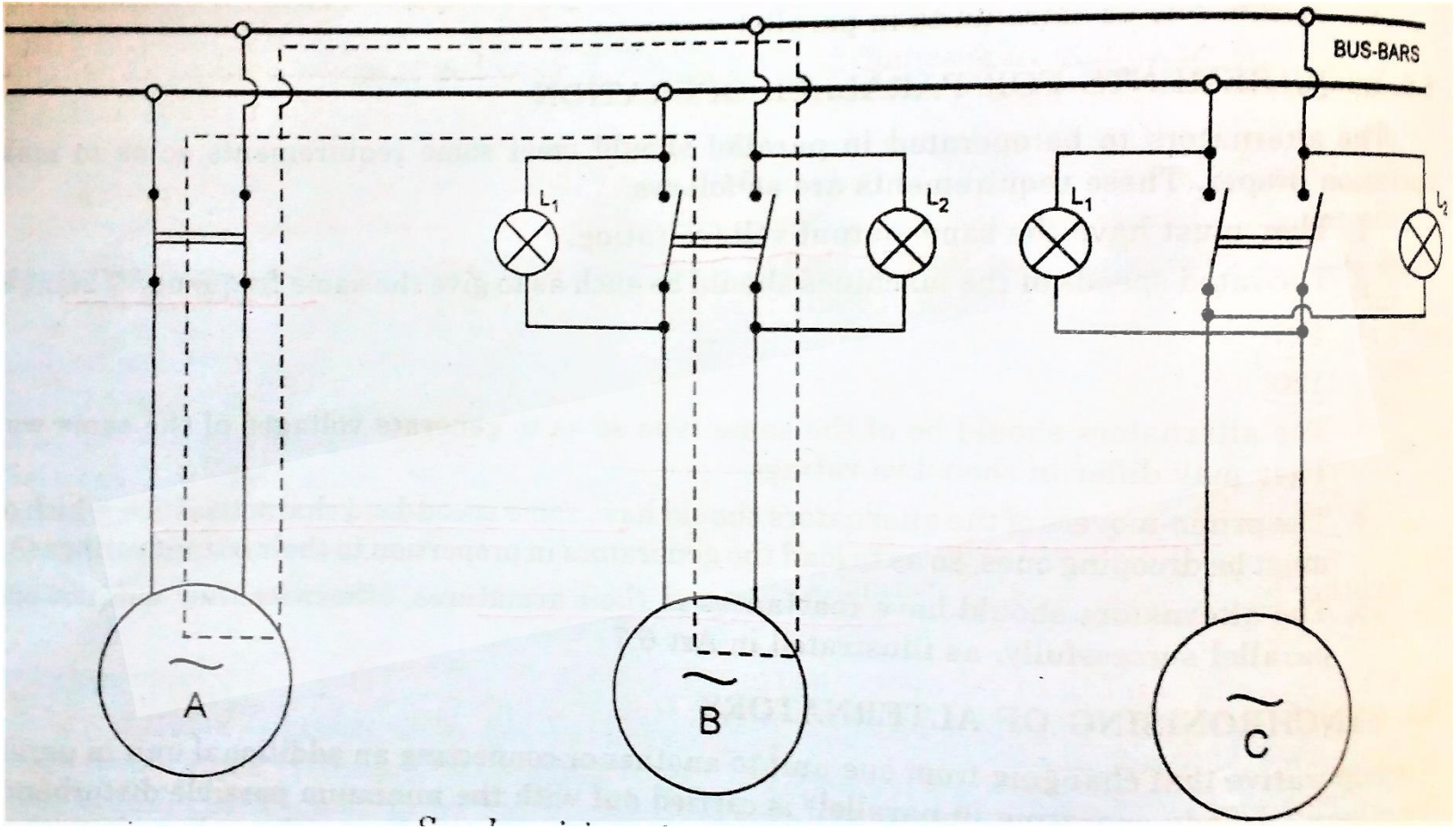


Figure 8 : Synchronising of single phase alternators

Dark Lamp Method

- Equality of terminal voltages can be determined by connecting voltmeters to the incoming and running alternators.
- Equality of frequency and phasing can be determined with the help of synchronising lamps.

Let alternator B is to be paralleled with alternator A,

- Prime mover of Alternator B is started and brought up close to the rated speed.
- Alternator B is excited and its voltage is raised by increasing the excitation.
- If frequencies of alternators A & B are exactly same and their terminal voltages w.r.t local series circuit → no resultant voltage will act across lamps L_1 and L_2 → Lamps will remain dark.
- If frequencies of A & B are not equal → flickering of lamps. Frequency of flickering = Difference of frequencies of two alternators.
- $\Rightarrow$ greater the difference in frequencies, greater will be the frequency of flick

Dark Lamp Method

- Synchronising is done at the middle of dark period.

  "At the time of synchronising of alternators, the speed of the incoming machine is adjusted until the lamps go in and out very slowly, the terminal voltage of the incoming alternator is made equal to bus bar voltage by adjusting the excitation of the incoming alternator and then switch of incoming alternator is closed in the middle of the dark period."

- But it is not so easy to judge the middle of dark period.

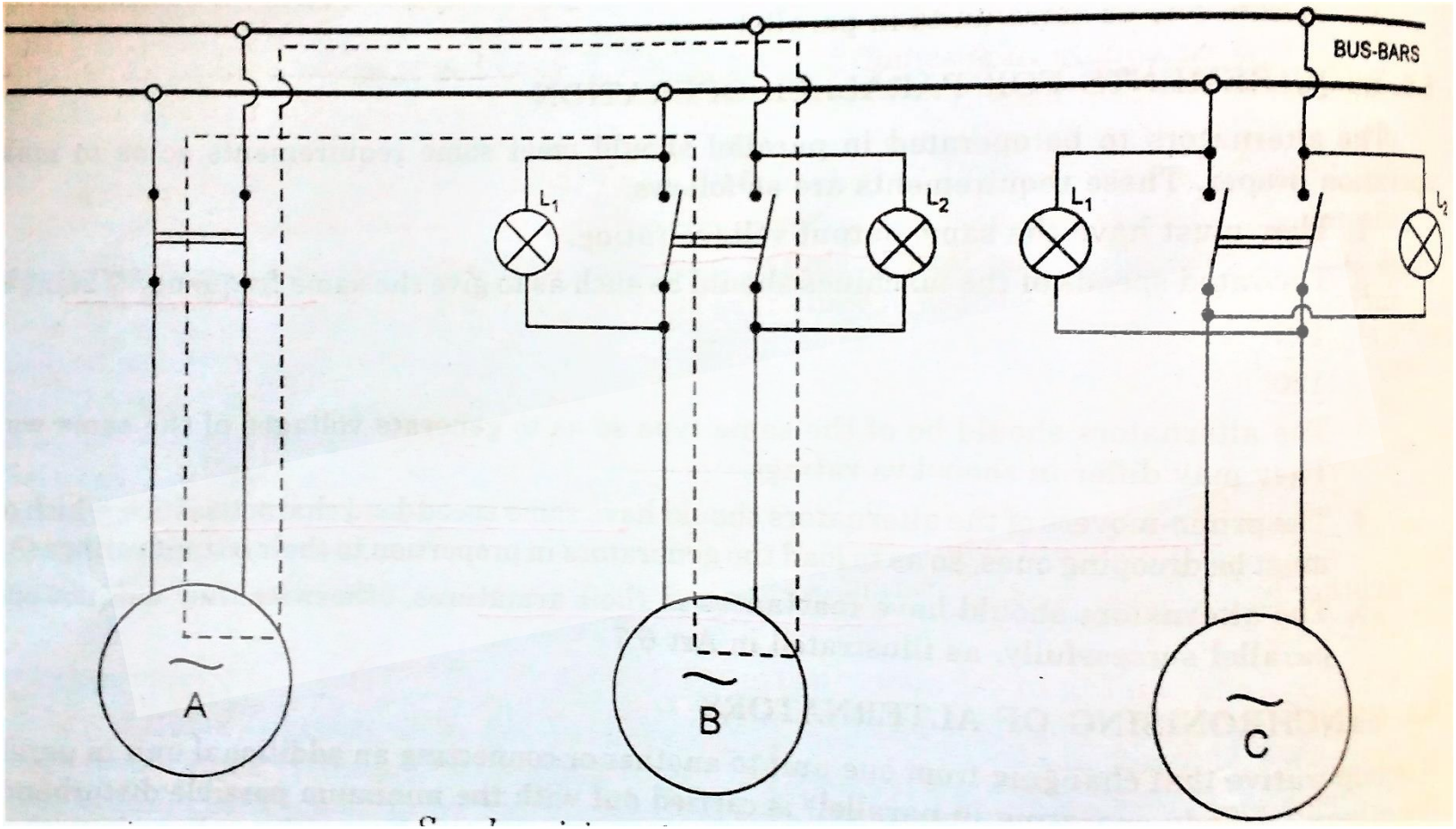


Figure 9 : Synchronising of single phase alternators

Bright Lamp Method

- Easy to judge the middle of bright period
- In **Bright lamp method**, lamps are cross-connected.
- Maximum voltage across lamps will occur when alternator C is in phase opposition with alternator A.
- Magnitude of maximum voltage that can exist across a lamp is double that of the voltage of one machine.
- After doing the adjustments, the synchronising switch is closed at the middle of bright period.

Synchronization of 3-Phase Alternators

"Phasing out the alternator" → The process of checking the phase sequence and getting it correct

1. Three Dark Lamp Method
2. Two Bright and One Dark Lamp Method
3. Using Synchroscope

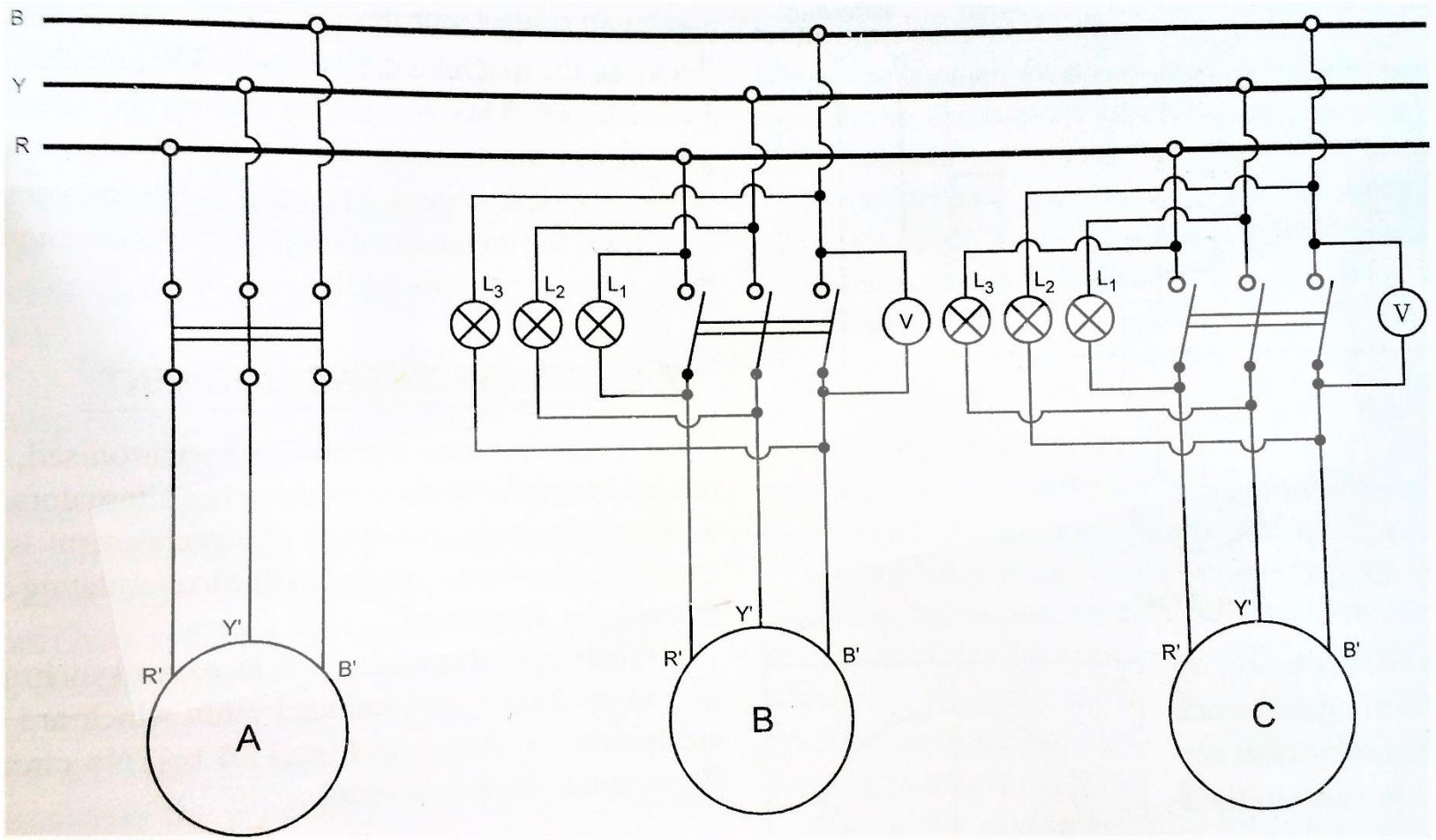


Figure 10 : Synchronising of 3-Phase Alternators

Three Dark Lamp Method

- Prime mover of alternator B is started and brought up close to the rated speed.
- Then alternator B is excited and its voltage is raised by increasing the excitation.
- If the incoming alternator B is properly connected, all the three lamps should become bright and dark together.
- If they bright and dim in sequence $\Rightarrow$ incoming alternator B is not properly connected with the bus-bars and the phase sequence of incoming alternator B must be reversed relative to the system.(Interchange any two leads on either the alternator side or the line side of the switch)
- The speed of incoming machine is further adjusted until the lamps flicker at a very rate and voltage is made equal to the bus bar voltage by adjusting the excitation.
- The synchronising switch is closed at the instant all the three lamps

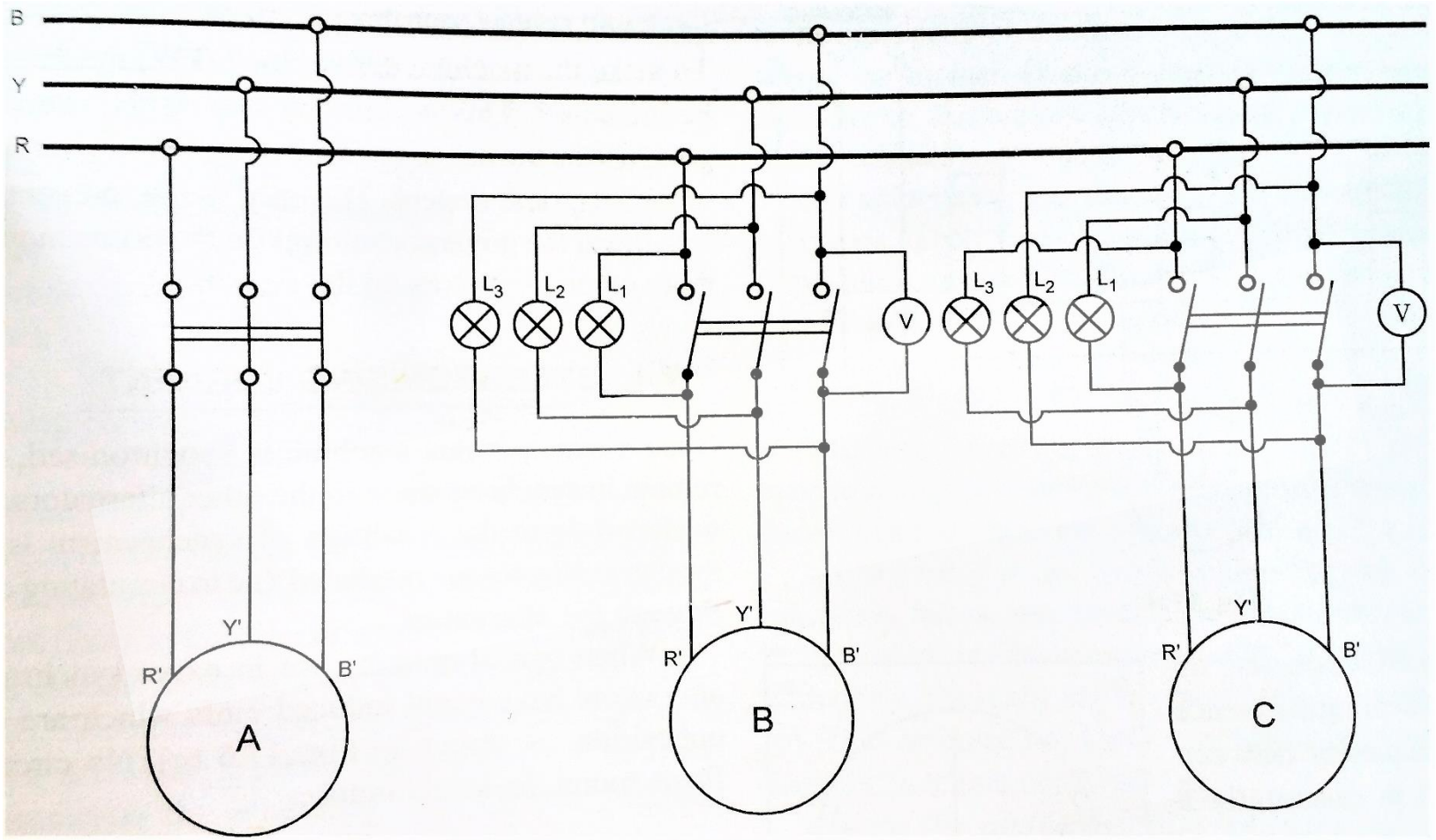
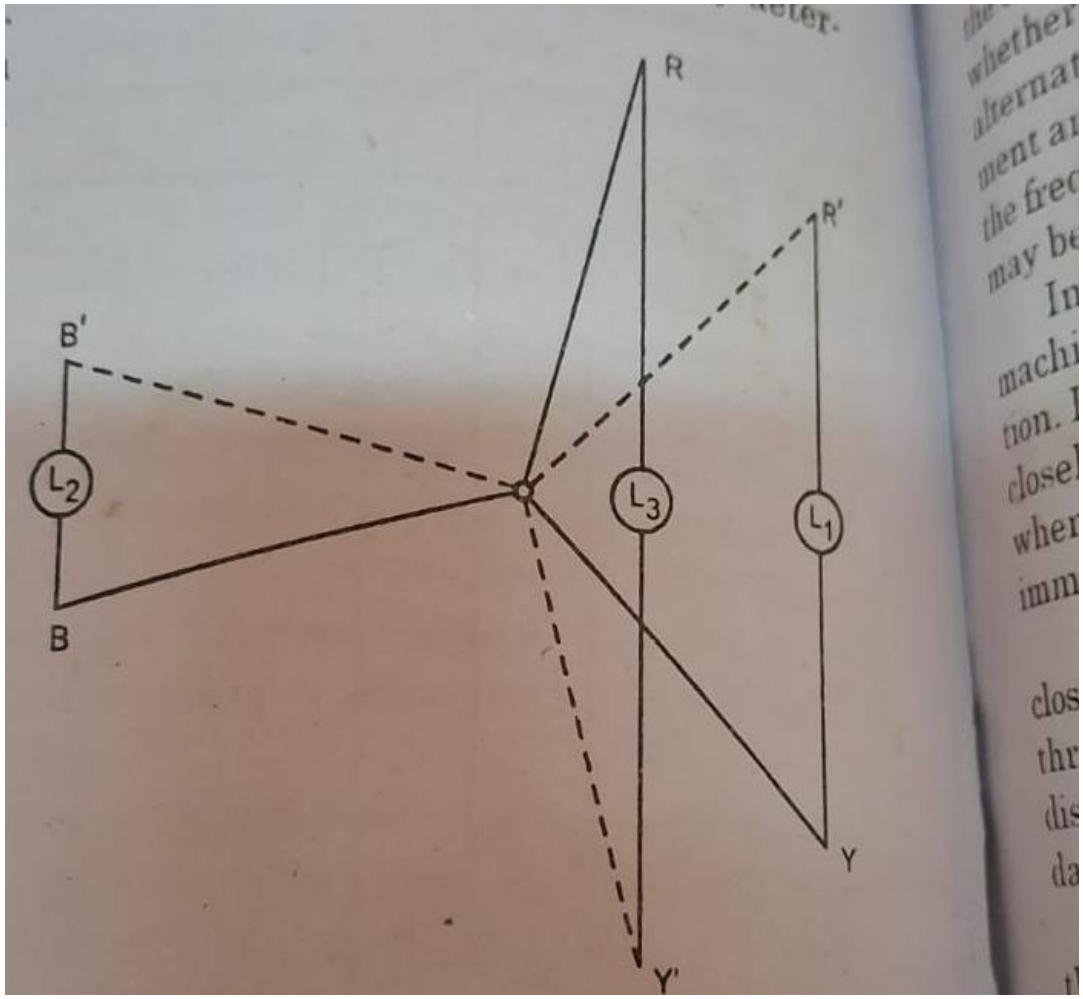


Figure 11 : Synchronising of 3-Phase Alternators

Two Bright and One Dark Lamp Meth

- When incoming alternator is in synchronism with running machine, lamps L_1 and L_3 are bright and L_2 is dark.
- Since near the point of synchronism, the brightness of one lamp increases and of another decreases. The instant at which the incoming machine is in synchronism with the bus-bars can be accurately determined and the paralleling switch will be closed at this instant.
- If the incoming machine is too fast → voltage across L_3 decreases, L_1 increases and L_2 decreases.
- If the incoming machine is too slow → voltage across L_3 increases, L_1 decreases and L_2 increases.
- Hence when the three lamps are placed in a ring, a light wave travelling in counter-clockwise direction indicates that the incoming machine is slow and light wave travelling in clockwise direction indicates that the incoming machine is fast.
- Synchronising switch is closed when changes in light are very slow and at the instant the lamp L is dark.


R
R'
B'
L2
L3
L1
B
Y
Y'

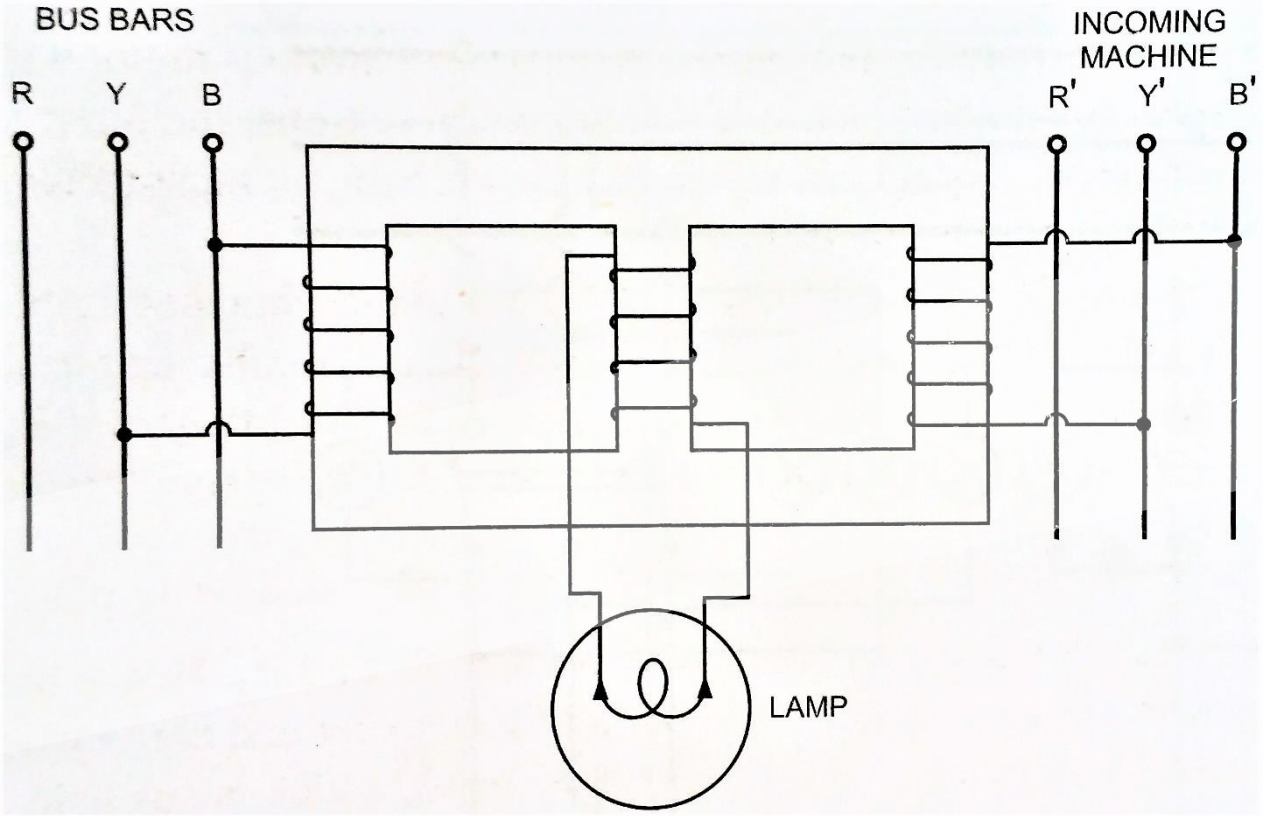


Figure 12 : Three Limbed Transformer and Lamp

Synchroscope

- Synchroscope → Instrument indicating the difference of phase and frequency between two voltages.
- Its a split-phase motor
- Torque is developed if the frequencies of the two voltages differ.
- Voltages from corresponding phases of incoming and running alternators are applie to synchroscope.
- When frequencies are equal, no torque is exerted on the pointer and the pointer stops at the middle of the dial.
- When pointerstops at vertical position, the frequencies are equal, the voltages are in phase and the paralleling switch can be closed.

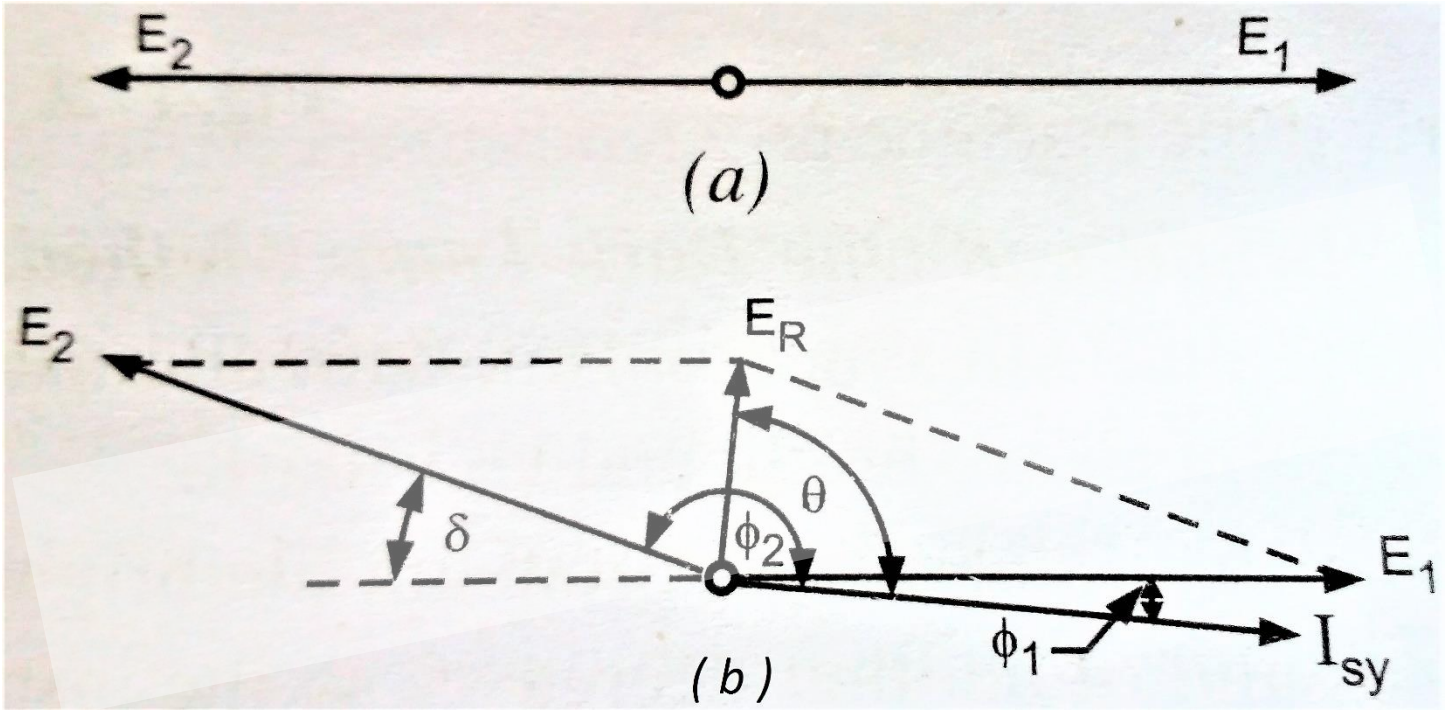


Figure 15 : Voltage Phasors

Synchronising Current

- When two alternators are in synchronism → have equal emfs in exact phase opposition → no circulating current (Fig. (a))
- When induced emfs are equal in magnitude but not in exact phase opposition → resultant emf around the local circuit /= zero → causes flow of current → **Synchronising current(I_{sy})**

  → Let alternator-2 tends to retard,
- E_2 falls back by a phase angle δ electrical degrees
- $|E_1| = |E_2| = E$
  but phase difference = 180 - δ

→ Resultant emf(E_R)

$$E_R = 2E\cos\frac{180 - \delta}{2}$$

$$= 2E\cos 90 - \frac{\delta}{2}$$

$$= 2E\sin\frac{\delta}{2}$$

$$= 2E \times \frac{\delta}{2}$$

$$= E\delta$$

→ Synchronising current(I_{sy})

$$\frac{E_R}{Z_{cs}} = \frac{E\delta}{Z_{cs}}$$

Synchronising Current

- I_{sy} lags behind E_R by an angle θ
- If $R_{ce} <<< X_{cs}$,

$$I_{sy} = \frac{E_R}{X_{cs}}$$

$\Rightarrow$ I_{sy} lags E_R by 90° $\Rightarrow$ almost in phase with E_1

- I_sy → generating current w.r.t machine 1 and motoring current w.r.t machine 2
- I_{sy} sets up synchronising torque(T_{sy}) which tends to accelerate machine no.2 and decelerate machine no.1
- Any departure from synchronism results in development of synchronising torque which tends to keep the machines in synchronism

Synchronising Power

- Power supplied by Machine no.1 = $E_1 I_{sy} Cos\Phi_1$
- Power received by Machine no.2 = $E_2 I_{sy} Cos\ (180 - \Phi_2)$
- Power supplied by Machine no.1 = Power received by Machine no.2 + copper losses
- Power supplied by Machine no.1 is called as **Synchronising power(P_{sy})**

$$P_{sy} = E_1 I_{sy} Cos\Phi_1$$

$$= E_1 I_{sy} \not{\delta}$$

$$= E \times \frac{E\delta}{X_{cs}}$$

$$= \frac{\delta E^2}{X_{cs}}$$

- Total synchronising power = $3P_{sy} = \frac{3\delta E^2}{X_{cs}}$

Synchronising Torque

$$3P_{sy} = T_{sy} \times \frac{2\pi N_s}{60}$$

Therefore Synchronising torque,

$$T_{sy} = \frac{3P_{sy} \times 60}{2\pi N_s}$$

where N_s is the synchronous speed in rpm

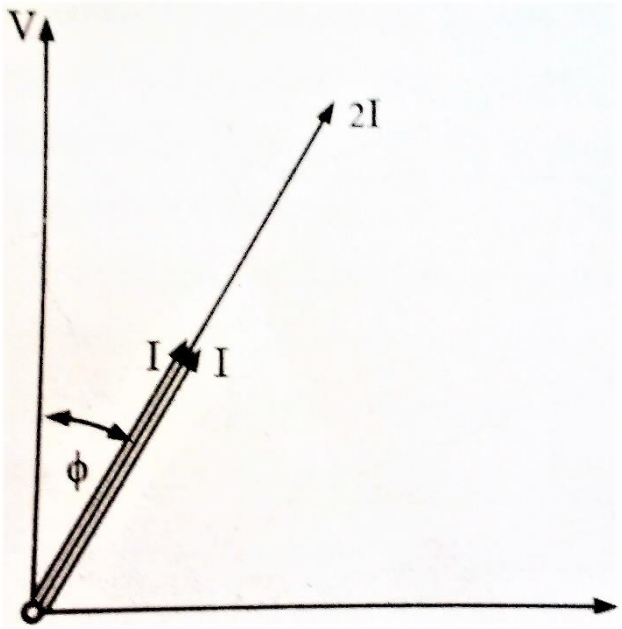


Figure 16 : (a)

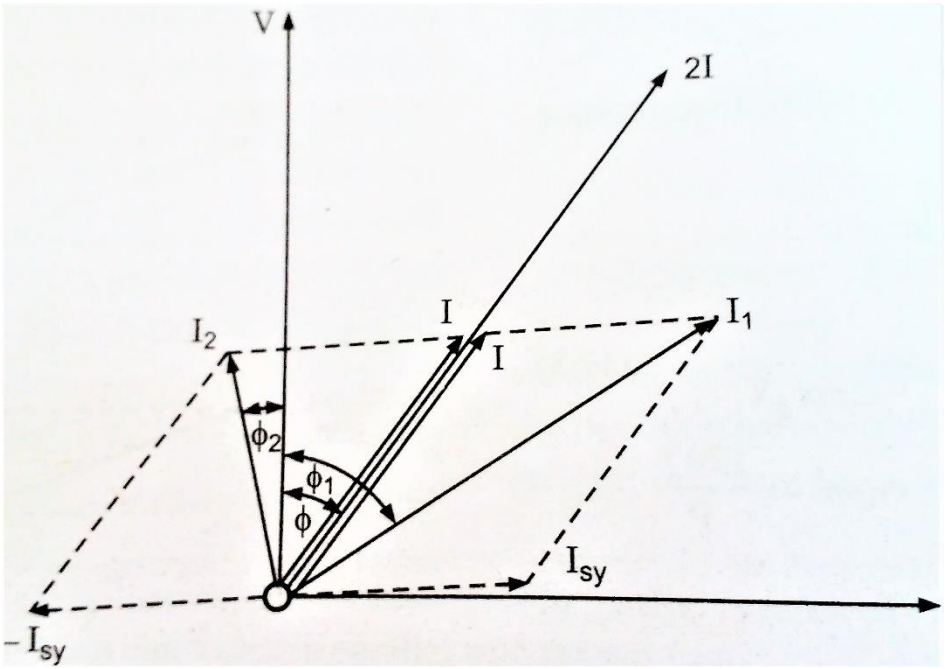


Figure 17 : (b)

MODULE III

Synchronous motor – Construction, Principle of operation, Effect of excitation on armature

Synchronous Motor - Features

- By construction it is very similar to synchronous generator
- It is a constant speed motor – speed decided by frequency
- Its power factor is controllable
- It requires both ac and dc supplies
  - AC for armature
  - DC for field excitation
- Synchronous motors are not self starting type
- Used for applications were constant speed operation is essential
  - Example: steel rolling mills, paper rolling mills

Construction of Synchronous motor

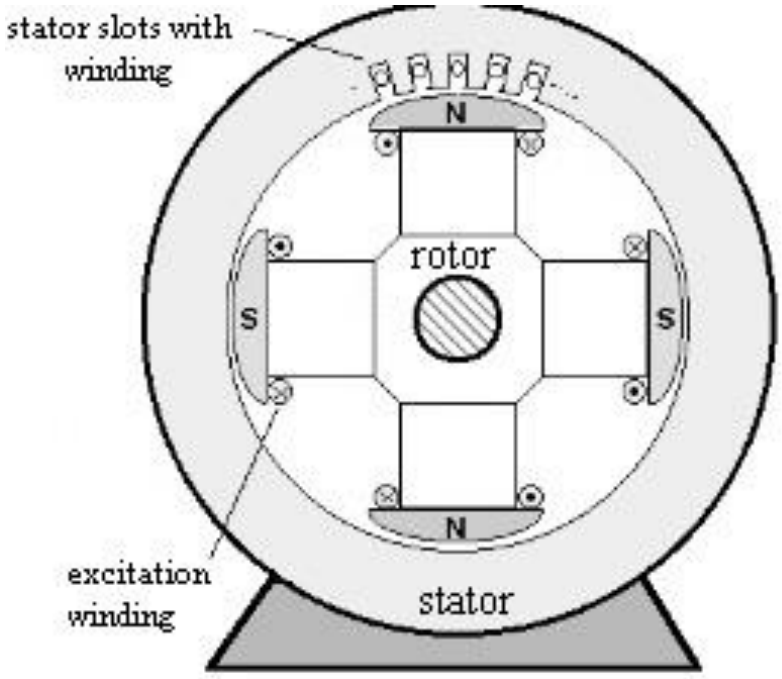
- It has mainly 2 parts
-

1. Stator
2. Rotor

1. Stator or Armature

- Stator is the stationary part
- Main parts of **stator** are ***stator core, stator winding & stator frame***
- ***Stator core*** contain an iron ring made of silicon steel laminations
- Slots are cut on the inner periphery of the stator core, in which stator conductors are placed
- Stator core is laminated to reduce eddy current loss
- axial & radial ventilating ducts to provide efficient cooling
- Slots can be open, semi closed or totally closed
- Open slots are commonly used

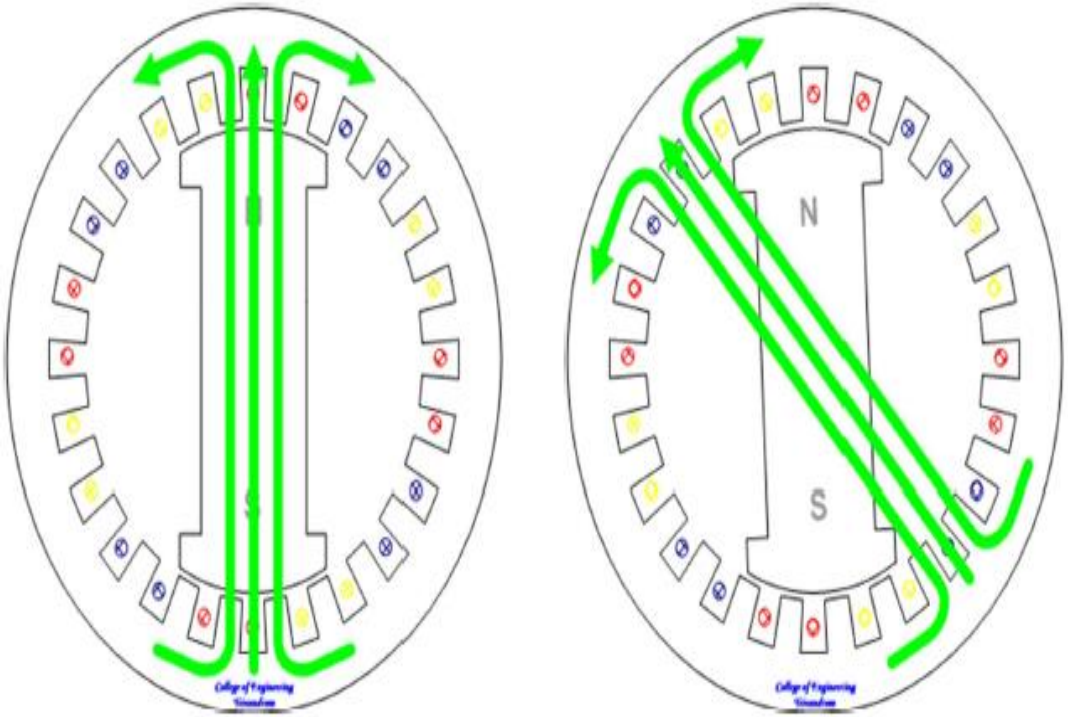
- Stator contain **a 3 phase winding** which is normally star connected
- Stator conductors are made of copper or aluminium
- Windings can be single layer or double layer, lap or wave, concentrated or distributed
- The whole structure is held in **a frame** made of cast iron or welded steel plate



2. Rotor or Field magnet system

- It is the rotating part of the machine
- Rotor contain Field magnets & are supplied with DC voltage
- For this external DC sources are used & exciting current is supplied to rotor through slip rings & brushes
- An exciter mounted on the motor shaft can also be used to provide necessary excitation.
- Normally, the rotor is having a salient pole structure
- Field poles contains laminated pole core & pole shoe (to reduce eddy current losses) and field windings

Why Synchronous Motors are not Self Starting ?

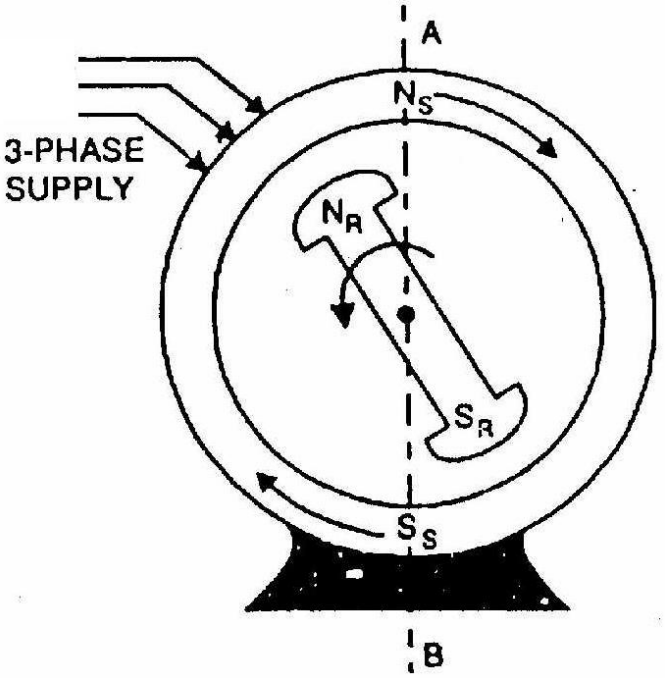


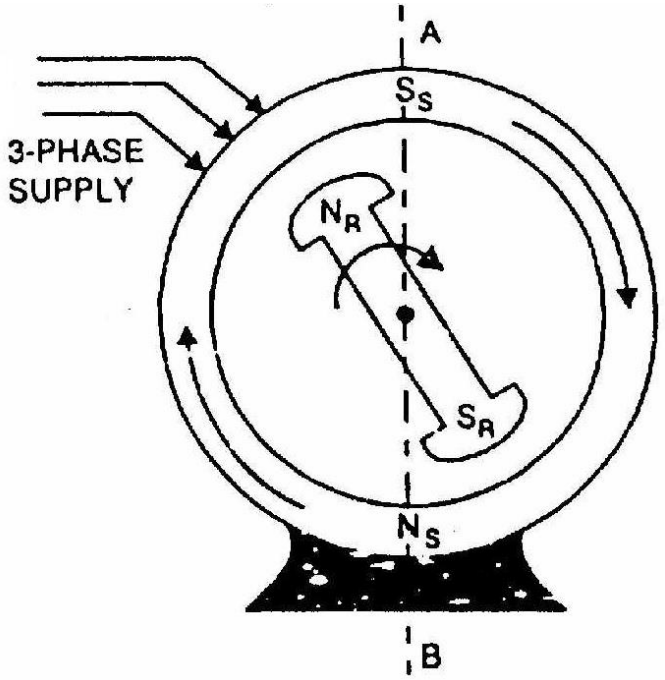
Due to the inertia of the rotor, it may not move quickly with the developed RMF when supply voltage is applied.

As the N and S poles of the RMF pass by, the rotor will tend to move in both clockwise and anticlockwise directions, but will not attain speed.

So synchronous motors are not self starting.

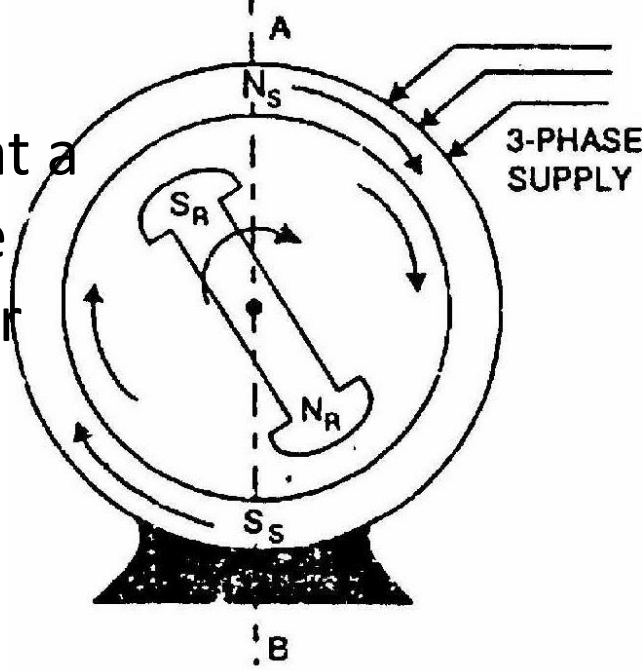
- Now stator North pole (Ns) & rotor North pole (Nr) will repel each other & rotor experiences a torque in anticlockwise direction
- After a period of half cycle, stator poles changes their position but rotor poles remain at previous position
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- Since stator poles changes their polarity rapidly, rotor experiences torque in opposite directions in a cycle.
- Due to high inertia of rotor, motor fails to start

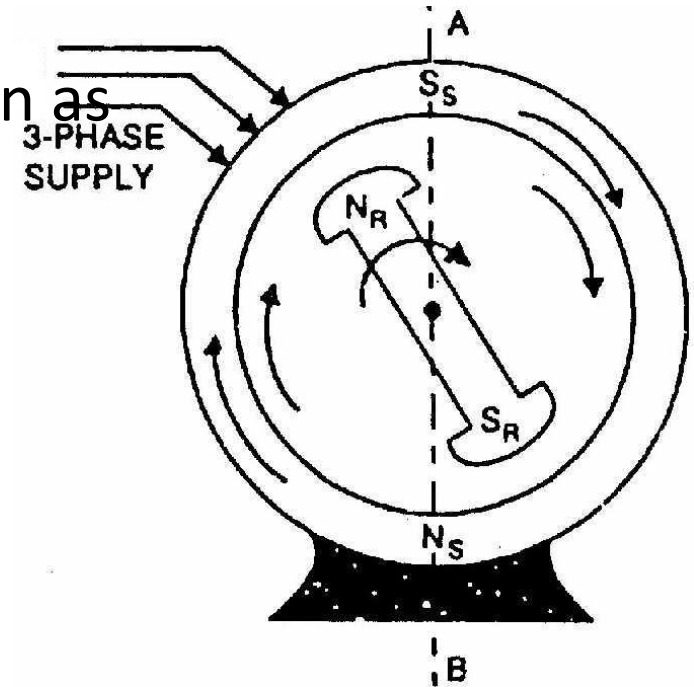




How to get unidirectional torque?

- If rotor is rotated by some external means at a speed near synchronous speed, in the same direction of stator magnetic field, then rotor will experience a unidirectional torque
- Now stator South pole (Ss) & rotor North pole (Nr) will attract each other & rotor experiences a torque in clockwise direction
- After a period of half cycle, stator & rotor magnetic field poles changes their position as shown in second figure
- rotor experiences a torque in clockwise direction.





- Under this condition, stator & rotor poles of opposite polarity experiences a strong attraction & the mutual attraction locks the rotor & stator together
- Now if the external prime mover is removed, rotor will continue to rotate at synchronous speed due to magnetic locking between stator & rotor

Starting methods of Synchronous motor

- Synchronous motor has no self starting torque & motor must first be brought upto synchronous speed by some external means for getting unidirectional torque in rotor.
- Various methods of starting of synchronous motors are

1. From DC source

- If a DC supply & a DC motor (shunt/compound) is available, the Synchronous motor is coupled to DC motor & is started using DC motor
- DC motor is coupled to Synchronous motor and dc motor is started & rotated at synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The DC motor is now disconnected from supply

2. By using an AC motor

- Here a small AC induction motor (called pony motor) is used for starting the Synchronous motor
- First the induction motor is started & will rotate Synchronous motor rotor at near synchronous speed
- AC supply is given to Synchronous motor (make sure that rotating magnetic field & rotor of synchronous motor are rotating in same direction)
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed
- The Induction motor is now disconnected from supply

3. By providing damper windings

- Synchronous motor can be made self starting by providing a special winding on rotor poles known as damper winding or squirrel cage winding
- Damper winding consist of short circuited copper bars placed in rotor poles
- When AC supply is given to stator, the synchronous motor will start as an Induction motor & rotates at a speed near synchronous speed
- Now DC field excitation is given & rotor gets locked to stator rotating magnetic field & continue to rotate at synchronous speed

Equivalent circuit of a Synchronous motor

- In a synchronous motor we are giving two supply, DC supply to rotor & 3 phase AC supply to stator
- When AC supply is given to stator, a current flows through stator windings
- Stator windings have resistance (R_a) & synchronous reactance (X_s)
- Also when a DC supply is given to rotor, a magnetic field is produced & it links stator conductors,
- As rotor rotates, flux linked with stator conductors changes & an EMF induces in stator windings. This is called back EMF (E_b) or counter EMF
- The back EMF opposes the stator supply voltage